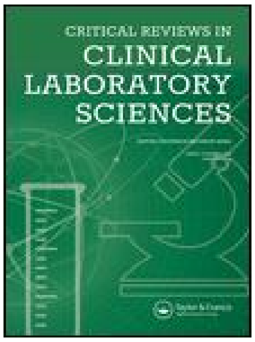

Critical Reviews in Clinical Laboratory Sciences

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/ilab20

The COVID-19 pandemic

Marco Ciotti , Massimo Ciccozzi , Alessandro Terrinoni , Wen-Can Jiang , Cheng-Bin Wang & Sergio Bernardini

To cite this article: Marco Ciotti , Massimo Ciccozzi , Alessandro Terrinoni , Wen-Can Jiang , Cheng-Bin Wang & Sergio Bernardini (2020): The COVID-19 pandemic, Critical Reviews in Clinical Laboratory Sciences, DOI: 10.1080/10408363.2020.1783198

To link to this article: https://doi.org/10.1080/10408363.2020.1783198

Published online: 09 Jul 2020.

https://doi.org/10.1080/10408363.2020.1783198

Taylor & Francis
Taylor & Francis Group

Check for updates

REVIEW ARTICLE

The COVID-19 pandemic

Marco Ciotti[a], Massimo Ciccozzi[b], Alessandro Terrinoni[c], Wen-Can Jiang[d], Cheng-Bin Wang[d] and Sergio Bernardini[c,e]

[a]Virology Unit, Tor Vergata University Covid-Hospital, Rome, Italy; [b]Unit of Medical Statistics and Molecular Epidemiology, University Campus Bio-Medico of Rome, Italy; [c]Department of Experimental Medicine, University of Tor Vergata, Rome, Italy; [d]Department of Laboratory Medicine, Chinese PLA General Hospital, Beijing, China; [e]Emerging Technologies Division, International Federation of Clinical Chemistry and Laboratory Medicine (IFCC), Milano, Italy

ABSTRACT

In December 2019, an outbreak of pneumonia of unknown origin was reported in Wuhan, Hubei Province, China. Pneumonia cases were epidemiologically linked to the Huanan Seafood Wholesale Market. Inoculation of respiratory samples into human airway epithelial cells, Vero E6 and Huh7 cell lines, led to the isolation of a novel respiratory virus whose genome analysis showed it to be a novel coronavirus related to SARS-CoV, and therefore named severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). SARS-CoV-2 is a betacoronavirus belonging to the subgenus *Sarbecovirus*. The global spread of SARS-CoV-2 and the thousands of deaths caused by coronavirus disease (COVID-19) led the World Health Organization to declare a pandemic on 12 March 2020. To date, the world has paid a high toll in this pandemic in terms of human lives lost, economic repercussions and increased poverty. In this review, we provide information regarding the epidemiology, serological and molecular diagnosis, origin of SARS-CoV-2 and its ability to infect human cells, and safety issues. Then we focus on the available therapies to fight COVID-19, the development of vaccines, the role of artificial intelligence in the management of the pandemic and limiting the spread of the virus, the impact of the COVID-19 epidemic on our lifestyle, and preparation for a possible second wave.

Abbreviations: ACE2: angiotensin converting enzyme 2; AI: artificial intelligence; ARDS: acute respiratory distress syndrome; CatL: cathepsin L; CRRT: continuous renal replacement therapy; CDC: Centers for Disease Control and Prevention; CI: confidence interval; CLIA: chemiluminescence assays; COVID-19: Corona virus disease 19; ECMO: extracorporeal membrane pulmonary oxygenation; ELISA: enzyme-linked immunosorbent assays; HCoV: human coronavirus; ICU: Intensive Care Unit; IL: interleukin; MERS-CoV: middle East respiratory syndrome coronavirus; POCT: point of care test; PCR: polymerase chain reaction; RBD: receptor binding domain; rRT-PCR: reverse real-time PCR; S protein: spike protein; SARS-CoV: severe acute respiratory syndrome coronavirus; SARS-CoV-2: severe acute respiratory syndrome coronavirus 2; TCM: traditional Chinese medicine; TMPRSS: surface transmembrane protease/serine protease; WHO: World Health Organization

ARTICLE HISTORY
Received 20 May 2020
Revised 7 June 2020
Accepted 12 June 2020

KEYWORDS
SARS-CoV-2; rRT-PCR; COVID-19; cytokine storm; pneumonia; ACE2; serology

1. Introduction

Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2), the seventh human coronavirus, was discovered in Wuhan, Hubei province, China, during the recent epidemic of pneumonia in January 2020 [1,2]. Since then, the virus has spread all over the world, and as of 20 May 2020, it has infected 4,806,299 people, and caused 318,599 deaths [3]. SARS-CoV-2 as well as SARS-CoV and Middle East respiratory syndrome coronavirus (MERS-CoV) cause severe pneumonia with a fatality rate of 2.9%, 9.6% and ∼36%, respectively [4–6]. The other four human coronaviruses, OC43, NL63, HKU1 and 229E, generally cause self-limited disease with mild symptoms [7].

In this review, we summarize the state of art of COrona VIrus Disease 19 pandemic (COVID-19, as defined by the World Health Organization [WHO] in February 2020), including the origin of SARS-CoV-2, its ability to infect human cells, epidemiology, clinical pathological and laboratory findings, molecular and serological diagnosis and safety issues. We also provide information on available therapies, vaccine development, and the potential role of artificial intelligence in

CONTACT Sergio Bernardini bernardini@med.uniroma2.it Department of Experimental Medicine, University of Tor Vergata, Via Montpellier 1, Rome, 00133, Italy

the governance of health care systems and its usefulness in fighting the COVID-19 outbreak.

2. Origin of SARS-CoV-2

Since the discovery of the novel coronavirus, SARS-CoV-2, scientists have debated its origin [8]. It has been speculated that SARS-CoV-2 is the product of laboratory manipulations. However, genetic data does not support this hypothesis and shows that SARS-CoV-2 did not derive from a previously known virus backbone [9].

Genomes analysis and comparison with previously known coronavirus genomes indicate that SARS-CoV-2 presents unique features that distinguish it from other coronaviruses: optimal affinity for angiotensin converting enzyme 2 (ACE2) receptor and a polybasic cleavage site at the S1/S2 spike junction that determines infectivity and host range [8,10].

SARS-CoV-2 is highly similar to bat SARS-like coronaviruses [2] and bat might be the reservoir host. RaGT13 is ~96% identical to SARS-CoV-2 with some differences in the spike receptor binding domain (RBD) that could explain the differences in ACE2 affinity between SARS-CoV-2 and SARS-like coronaviruses.

The polybasic cleavage site of SARS-CoV-2 is not present in pangolin beta-coronavirus, which share similarities with SARS-CoV-2. Also, the sequence of RBD of the spike protein (S) suggests that it arose from a natural evolutionary process [8].

Estimates of the most recent common ancestor of SARS-CoV-2 date the epidemic to between late November 2019 and the beginning of December 2019, which is compatible with the first reported cases [11]. Thus, there was unnoticed human transmission after the zoonotic event and before the acquisition of the polybasic furin cleavage site [8].

3. Epidemiology

3.1. Disease presentation

Patients with SARS-CoV-2 infection may present symptoms ranging from mild to severe with a large portion of the population being asymptomatic carriers. The most common reported symptoms include fever (83%), cough (82%) and shortness of breath (31%) [12]. In patients with pneumonia, chest X-ray usually shows multiple mottling and ground-glass opacity [12,13].

Gastrointestinal symptoms such as vomiting, diarrhea, and abdominal pain are described in 2–10% of the patients with COVID-19 [12,14], and in 10% of patients, diarrhea and nausea precede the development of fever and respiratory symptoms [12].

COVID-19 patients usually show decrease lymphocyte and eosinophils counts, lower median hemoglobin values as well as increases in WBC, neutrophil counts, and serum levels of CRP, LDH, AST, and ALT [15]. Moreover, initial CRP serum levels have been reported to be an independent predictor for the development of severe COVID-19 infection [16,17].

Although the main target of coronavirus infection is the lung, the wide distribution of ACE2 receptors in organs [18] may lead to cardiovascular, gastrointestinal, kidney, liver, central nervous system and ocular damage that has to be closely monitored [19].

The cardiovascular system is often affected, with complications including myocardial injury, myocarditis, acute myocardial infarction, heart failure, dysrhythmias, and venous thromboembolic events, and monitoring with high sensitivity cardiac troponin may be useful [20].

Patients presenting with acute respiratory distress syndrome may worsen rapidly and die of multiple organ failure [12] induced by the so-called "cytokine storm".

Indeed, a cytokine profile resembling the secondary hemophagocytic lymphohistiocytosis syndrome has been described in severe COVID-19 cases, and is characterized by increased interleukin (IL)-2, IL-7, granulocyte colony stimulating factor, interferon-γ inducible protein-10, monocyte chemoattractant protein 1, macrophage inflammatory protein 1-α, and tumor necrosis factor-α [11]. In addition, elevated levels of ferritin and IL-6 are predictors of fatality, and death is likely due to hyperinflammation induced by the virus [21]. Based on this evidence, tocilizumab (IL-6 receptor blockade) is administered to patients with COVID-19 pneumonia and elevated serum IL-6 to reduce inflammation in the lungs.

Elevation of D-dimer levels has been associated with the severity of COVID-19. Subjects with severe COVID-19 have significantly higher values of D-dimer than those without (weighted mean difference 2.97 mg/L; 95% CI: 2.47–3.46 mg/L) [22]. The elevated D-dimer levels may reflect the risk of disseminated coagulopathy in patients with severe COVID-19, which may require anticoagulant therapy [22].

The Italian Agency of Medicine (AIFA) has recently approved a clinical trial (INHIXACOVID19 study) in which enoxeparin is given subcutaneously to patients with COVID-19 to prevent thromboembolism-related complications. Heparin also has antiviral activity. It is known for its ability to prevent viral infection including coronaviruses infection. Indeed, heparin has a structure similar to that of heparan sulfate that is present on mammalian cellular surfaces and that is utilized by coronaviruses to enter cells [23,24]. In the presence of heparin, the interaction of the S protein with heparan sulfate may be blocked, thus preventing cell entry.

Furthermore, heparin may inhibit proteases involved in virus infectivity [25]. Indeed, SARS-CoV-2 entry requires the cleavage of the S1–S2 subunits followed by the fusion of S2 to the cell membrane. The latter requires the action of host proteases such as cathepsins, cell surface transmembrane protease/serine proteases (TMPRSS), furin, trypsin and factor Xa that are inhibited by heparin.

The course of COVID-19 disease in children is generally asymptomatic or mild compared to that seen in adults, for reasons that are yet to be clearly elucidated. Nonetheless, severe and fatal cases have been reported in children. Clinical laboratory data in children is quite different from adults as reported in a recent meta-analysis that showed an inconsistent alteration of the leukocyte index [26]; however, elevations in the levels of CRP, procalcitonin and LDH were also found in children with severe disease. Interestingly, creatine kinase-MB was elevated in one-third of patients and this raised the suspicion of cardiac involvement in COVID-19 pediatric patients, as recently reported [27].

3.2. SARS-COV-2 transmission

As with other respiratory viruses, SARS-CoV-2 transmission occurs with high efficacy and infectivity mainly through the respiratory route. Droplet transmission is the main recognized route, although aerosols may represent another important route [28,29]. Estimates of the reproduction number (R0) of SARS-CoV-2 range from 1.4 to 2.5 [Statement on the meeting of the International Health Regulations (2005) Emergency Committee regarding the outbreak of novel coronavirus (2019-nCoV), https://www.who.int/news-room/detail/23-01-2020-statement-on-the-meeting-of-the-international-health-regulations-(2005)-emergency-committee-regarding-the-outbreak-of-novel-coronavirus-(2019-ncov)] to 2.24–3.58 [30].

Similar to SARS-CoV, the oral-fecal route may be another route of transmission of the virus. SARS-CoV-2 RNA has been detected in the stool of patient with COVID-19 pneumonia [31]. Therefore, sewage may have a role in the transmission of SARS-CoV-2. In light of that, technical treatment such as biosorbents capable of retaining and inactivating the virus should be considered [32].

SARS-CoV-2 has been detected in saliva of infected individuals [33]; this can be attributed to the presence of ACE2 receptors in epithelial cells lining the salivary gland ducts [34].

In some studies, patient urine has been tested for SARS-CoV-2 viral RNA. Amongst these studies, the pooled rate of RNA positivity was about 5–6%; nevertheless, the duration of viral shedding in urine samples as well as the infectivity of urine remains to be established [35].

SARS-CoV-2 RNA has also been detected on inanimate surfaces such as door handles and the surface of cell phones in residential sites of patients with confirmed COVID-19. Thus, individuals who have come into contact with infected surfaces could be infected if they touch their eyes, mouth or nose [29].

The vertical transmission of SARS-CoV-2 is debated; a series of nine pregnant women with confirmed COVID-19 showed no mother to child transmission. In addition, SARS-CoV-2 was not detected in breast milk, indicating that the virus cannot be transmitted with breastfeeding [36]. Nevertheless, a newborn with elevated IgM against SARS-CoV-2 born to a mother with COVID-19 has been recently reported. IgM antibodies along with IgG antibodies were detected 2 h after delivery. Il-6 and IL-10 were also elevated, while polymerase chain reaction (PCR) performed on consecutive nasopharyngeal swabs from 2 h to 16 days of age was always negative. Considering that IgM cannot cross the placenta and be transferred to the fetus, it could be hypothesized that the infant was infected *in utero* even if amniotic fluid was not tested for SARS-CoV-2 RNA [37].

Finally, the eyes may be a route of transmission of SARS-CoV-2. SARS-CoV-2 RNA was detected in ocular swabs of a patient with confirmed COVID-19 3 days after onset of symptoms and at 27 days when a nasopharyngeal swab tested negative by PCR. Interestingly, the virus from an ocular swab was propagated in Vero E6 cells, suggesting that ocular secretions could be infectious [38]. Although no conclusive data is available, goggles should be worn when examining patients with suspected or confirmed COVID-19 [39].

3.3. SARS-CoV-2 incubation period

Determination of the incubation period of SARS-CoV-2 infection is crucial for determining the duration of quarantine, to evaluate the efficacy of entry screening and contact tracing. Based on a Weibull distribution, it was estimated that the mean incubation period is 6.4 days (95% confidence interval (CI): 5.6–7.7), with a range of 2.1–11.1 days (2.5th–97.5th percentile) [40]. Similar estimates have been made by other authors. In a study by Lauer et al. [41], it was estimated that the median incubation period was 5.1 days (95% CI, 4.5–5.8 days), and that 97.5% of infected individuals would develop symptoms within 11.5 days (CI, 8.2–15.6 days) of infection. Therefore, the 14-day period of active monitoring recommended by health authorities is justified by the evidence [42,43]. Longer monitoring can be required in particular cases. It was estimated that 101 out of every 10,000 cases (99th percentile, 482) may develop symptoms after 14 days of active monitoring or quarantine [41].

4. Viral testing

4.1. Reverse real-time PCR assays

Suspected cases of SARS-CoV-2 infection are confirmed by detection of specific and unique viral sequences using a reverse real-time PCR (rRT-PCR) assay. Immediately after the declaration by the Chinese Health Authorities, on 7 January 2020, that the pneumonia outbreak in Wuhan was caused by a novel coronavirus, a European network of laboratories developed an rRT-PCR protocol based on the alignment and comparison of available bat-related coronavirus and SARS-CoV genome sequences plus five sequences from the novel coronavirus SARS-CoV-2 that were released by the Chinese authorities [44]. Three rRT-PCR assays were developed. The first line assay targets the *E* gene common to the coronaviruses belonging to *Sarbecovirus* subgenus and encoding the envelope protein. The second assay targets the *RdRp* gene encoding the RNA-dependent-RNA-polymerase. This assay contains two molecular probes: one reacts with the SARS-CoV and SARS-CoV-2 *RdRp* gene, while the second one (RdRP_SARSr-P2) reacts with SARS-CoV-2 *RdRp* gene. The third assay targets the *N* (nucleocapsid) gene. This protocol was adopted in 30 European countries [45].

Recently, a new PCR protocol targeting a different region of the *RdRp/Hel* gene showed a higher sensitivity and specificity than the RdRP_SARSr-P2 assay [46].

The US Centers for Disease Control and Prevention (CDC) protocol targets the *N* gene of SARS-CoV-2. Two primer/probe sets directed toward different regions of *N* gene were selected. In addition, a primer/probe set that detects the human RNase *P* gene in control samples and clinical specimens is included (https://www.fda.gov/media/134922/download. Revision 3, 30 March 2020).

Although amplification tests are sensitive, some infections are missed. The reasons for this may be the quality of the collected specimen, time of collection (very early phase of infection or too late during infection), viral load below the limit of detection of the assay, incorrect handling of the specimen or shipping issues. In the case of low viral load in the upper respiratory tract, a deeper specimen may be required to make a diagnosis [47]. Indeed, in SARS and MERS patients, viral RNA in the upper respiratory tract peaked in the first 7–10 days after symptoms onset, while in the lower respiratory tract, viral RNA was still detected 2–3 weeks after disease onset [47,48]. Repeat testing can also be performed in the case of nasopharyngeal swabs initially being negative as this increases the chance of detecting SARS-CoV-2 in the nasopharynx [49].

Several real-time PCR assays obtained the CE mark for *in vitro* diagnostics and are available on the market. Table 1 summarizes the gene targets, the technical features of the assays and the validated specimen types of the real-time PCR assays cleared by the Italian Ministry of Health.

Point of care tests (POCT) such as Xpert® Xpress SARS-CoV-2 (Cepheid, Sunnyvale, CA, USA) QIAstat-Dx Respiratory 2019-nCoV Panel (QIAGEN, Hilden, Germany), and Simplexa™ COVID-19 Direct kit (DiaSorin Molecular LLC, Cypress, CA, USA) deliver results in about 30-60 min. They do not require skilled technicians and the hands-on time is less than 1 min. These tests can be very useful when clinicians have to make rapid treatment decisions.

A sensitive rRT-PCR assay could be performed on pooled samples. In the present situation where

Table 1. Real-time PCR assays cleared by the Italian Ministry of Health for detection of SARS-CoV-2.

Manufacturer	Specimen type	Method	Gene target	Sensitivity
Bosphore Novel Coronavirus (2019-Ncov) Detection Kit	Nasopharyngeal swab, oropharyngeal swab, sputum, bronchoalveolar lavage	Fluorescence RT-PCR	E, orf1ab	25 copies/reaction
STANDARD M nCoV Real-Time Detection Kit	Nasopharyngeal swab and throat swab, sputum	Fluorescence RT-PCR	E, orf1ab	NA*
Allplex 2019-nCoV assay	Sputum, nasopharyngeal swab, nasopharyngeal aspirate, bronchoalveolar lavage, throat swab	Fluorescence RT-PCR	E, RdRp, N	100 copies/reaction
QUANTY COVID-19	Nasopharyngeal swab, oropharyngeal swab, sputum, serum	Fluorescence RT-PCR	N	NA*
GENEFINDER COVID-19 PLUS REALAMP KIT	Bronchoalveolar lavage fluid, throat swab, sputum	Fluorescence RT-PCR	E, RdRp, N	10 copies/reaction

*NA: not available.

shortage of reagents can be a problem, sample pooling can be a valid alternative to allow the screening of a large number of people in a short time frame. In a recent study, a positive single sample could be detected in a pool of up 32 samples with an estimated false negative rate of 10% [50]. Pooled screening could be implemented to detect SARS-CoV-2 in the community [50,51].

A recent paper showed that in confirmed COVID-19 patients, saliva may be a more sensitive specimen for SARS-CoV-2 detection than nasopharyngeal swab. The authors also reported less variability in self-sample collection of saliva compared to nasopharyngeal swabs. This observation could open the way to at-home self-administered sample collection for large-scale screening of SARS-CoV-2 [52].

The use of urine for diagnostic purposes is still the object of debate. The receptor ACE2, which is used by SARS-CoV-2 to infect human cells, is present not only in the respiratory tract but also in the urogenital system: renal proximal tubule cells, bladder urothelial cells, Leydig cells and cells in the testicular seminiferous ducts in testis [53]. Indeed, patients with COVID-19 may present with kidney damage (proteinuria, elevated serum creatinine, high urea) or severe acute kidney failure. Damage to the reproductive system may occur as well [53]. Taken together, these data suggest that the urogenital system may represent a route of transmission of SARS-CoV2. Some authors reported the identification of the virus in urine [54]. Nevertheless, to date, there is insufficient evidence that urine can be used as a biological specimen for COVID-19 diagnosis.

4.2. CRISPR-Cas12-based lateral flow assay for detection of SARS-CoV-2

Recently, a CRISPR-Cas12-based assay called SARS-CoV-2 DNA endonuclease-targeted CRISPR Trans Reporter (DETECTR) has been developed for the diagnosis of SARS-CoV-2 infection. This assay performs simultaneous reverse transcription and isothermal amplification of RNA extracted from nasopharyngeal or oropharyngeal swabs using loop-mediated amplification (RT-LAMP) [55], followed by Cas12 detection of coronavirus sequences. Detection of the virus is confirmed by cleavage of a reporter molecule. The assay targets the *E* and *N* regions but unlike the CDC assay, this one does not target the *N1* and *N3* regions. Amplification of the *E* region allows the identification of three SARS-like coronaviruses [SARS-CoV-2 (accession NC_045512), bat SARS-like coronavirus (bat-SL-CoVZC45, accession MG772933) and SARSCoV (accession NC_004718)], whereas the *N* region specifically detects SARS-CoV-2.

The DETECTR assay can be run in 30–40 min and is visualized on a lateral flow strip. The test is positive if both *E* and *N* genes are detected or presumptive positive if either *E* or *N* gene is detected. The limit of detection (LOD) of this DETECTR assay is 10 copies/μl *vs* 1 copy/μl for the CDC assay. The positive predictive agreement and the negative predictive agreement of DETECTR assay versus the CDC assay were 95% and 100%, respectively [56].

4.3. Viral load in respiratory samples

Viral load determination performed on nasopharyngeal swabs demonstrated that mild clinical cases had a lower viral load in their respiratory specimens compared with severe cases. The mean viral load in severe cases was 60-fold higher than mild cases. Stratification of the data in relation at the time of sampling after disease onset showed that delta cycle threshold (delta Ct) values were significantly lower in severe cases than mild cases in the first 12 days of disease. In mild cases, viral clearance occurred earlier and after 10 days, 90% of the patients repeatedly tested negative by PCR. By contrast, PCR was still positive at day 10 or beyond in all severe cases. These preliminary data suggest that determination of SARS-CoV-2 load may be useful for monitoring the patients with COVID-19 disease and for predicting prognosis and assessing disease severity [57].

5. Serology

Several serological assays, including enzyme-linked immunosorbent assays (ELISA), chemiluminescence assays (CLIA), rapid antibody tests, and western blotting, have been developed since the beginning of the SARS-CoV-2 pandemic. The ELISA test developed by Wantai Biological Pharmacy Enterprise Co. (Beijing, China) detects total antibody, IgM and IgG against SARS-CoV-2 [58]. Total antibodies were detected based on a double-antigen sandwich immunoassay using recombinant antigens containing the RBD of the S protein of SARS-CoV-2 as the immobilized antigen and horse radish peroxidase (HRP) as the conjugated antigen. The IgM μ-chain capture method was used to detect IgM antibodies (IgM-ELISA), using the same HRP-conjugate RBD antigen as in the double-antigen sandwich immunoassay. The IgG antibodies were detected by an indirect ELISA test (IgG-ELISA) based on the recombinant nucleoprotein antigen. The specificity of the assays was determined by testing plasma samples of healthy individuals collected before the SARS-CoV-2 outbreak, and was 99.1%, 98.6% and 99.0% for total antibody, IgM and IgG, respectively [58]. Comparing the results obtained by real-time PCR with those generated by the antibody assays in the first week of illness, PCR showed a higher sensitivity than antibodies assays, 66.7% *vs* 38.3%. However, from days 8 to 12, the sensitivity of the antibody tests overtook that of the RNA test, and in the late phase of disease, the sensitivity of the antibody tests increased even further compared to the RNA test [58]. Nevertheless, to date, all the international professional organizations, the US Food and Drug Administration and the CDC do not recommend that serology tests be used for diagnosis.

Other research groups developed ELISA assays that used as antigens a modified full-length S protein and the RBD of SARS-CoV2. Using this approach, COVID-19 seroconverters were identified as early as 3 days post symptom onset. Similarly, an antibody test designed to detect E and N antigens detected IgM within one week from disease onset. IgM was detectable for about a month and then gradually disappeared, while IgG was detected after 10 days and was detected for a longer period of time [59].

Using a magnetic chemiluminescence enzyme immunoassay, 100% of patients with COVID-19 were found to be IgG positive 19 days after symptom onset. The median day for both IgG and IgM seroconversion was 13 days post symptoms. Seroconversion for IgM and IgG occurred simultaneously or sequentially. Three groups of patients were identified: patients with synchronous seroconversion of IgG and IgM, patients with IgM seroconversion earlier than IgG seroconversion, and patients with IgG seroconversion earlier than IgM seroconversion. IgM and IgG plateaued 6 days after the first determination. Interestingly, screening of close contacts of patients with COVID-19 showed that few individuals with negative RT-PCR and no symptoms tested positive to IgG and/or IgM, confirming that serology can help in obtaining better estimates of the spread of SARS-CoV-2 [60].

In the case of coronaviruses infection, neutralizing antibodies are elicited by the RBD region of the S protein [61,62]. This subset of antibodies is particularly important because they can prevent the entry of the virus into the cell by binding to epitopes on the surface of the viral particle and blocking their interaction with the receptor. Neutralizing antibodies were first detected in the serum of a hospitalized Chinese female tourist between day 4 and 9 from onset of symptoms [61].

In the past, a cocktail of neutralizing antibodies has been used in the treatment of SARS-CoV and Ebola virus infections. The combination of neutralizing antibodies was more effective than a single antibody [63,64]. It is likely that a similar approach will be undertaken in the treatment of SARS-CoV-2 infection. A recently developed human monoclonal antibody, 47D11, against SARS-CoV-2 is able to neutralize the infection of Vero E6 cells by SARS-CoV and SARS-CoV-2 *in vitro* [65]. The capacity of 47D11 to cross-react with SARS-CoV and SARS-CoV-2 suggests that the antibody likely targets the conserved structure of the $S1_B$ RBD [65]. It is important to know whether this antibody can be useful in the development of serological assays for SARS-CoV-2 or antigen detection assays. At the clinical level, the 47D11 monoclonal antibody could represent a new therapeutic option for treating COVID-19 disease or for preventing infection.

Many in-house ELISA assays that are able to detect antibodies against S1, RBD and N of SARS-CoV-2 have been developed, although cross-reaction with SARS-CoV was observed [66] due to the high degree of similarity between the S1 and RBD proteins of the two coronaviruses as well as between the N proteins (90% similarity). However, it should be pointed out that SARS-CoV antibodies have been reported to have waned in the 17 years since the SARS-CoV epidemic, making it unlikely that antibodies against SARS-CoV could still be detectable in the population and create false positive results. Moreover, a study performed 6 years after the SARS epidemic showed that 91% of the serum samples of the patients previously infected by SARS-CoV were negative for specific IgG [67].

Detection of neutralizing antibodies against SARS-CoV-2 is also important for identifying individuals who mount a strong immuneresponse against the virus and whose serum/plasma could be used therapeutically to fight SARS-CoV-2 infection.

Major IVD companies [68] have introduced SARS-CoV-2 serology tests and offer IgA, IgG, IgM or mixed assays on automated immunoassay analyzers in the clinical laboratory.

Considering the numerous immunoassays (ELISA and CLIA) already developed by several research groups and those that will become available in the near future, it is mandatory to standardize the assays using recognized reference standards or at least to harmonize them. Until standardization/harmonization are realized, it would be appropriate that each laboratory calculate its own cutoff using receiver operating characteristic curves. In many countries, serological assays are also in

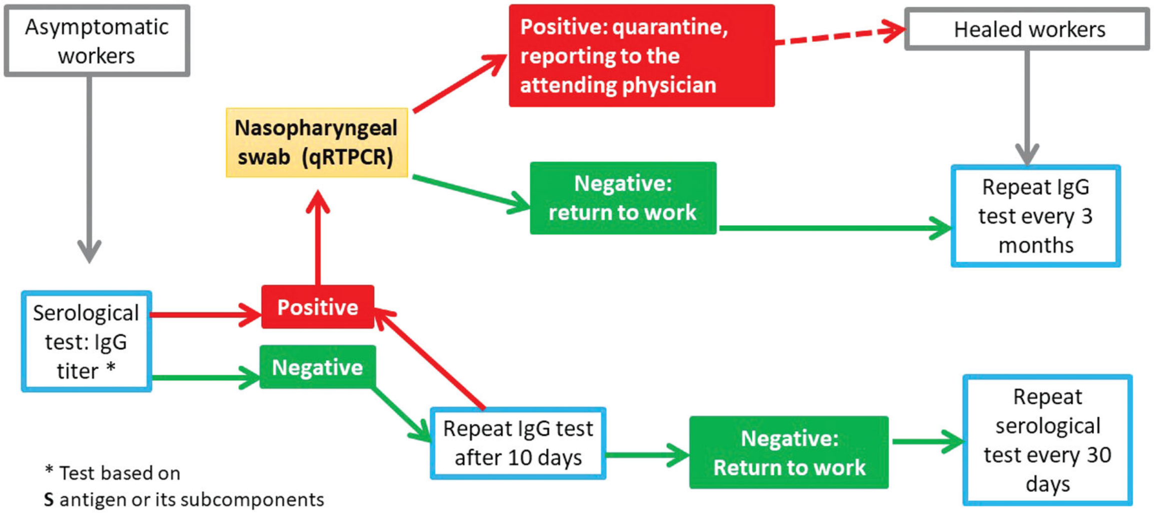

Figure 1. Flowchart proposal to overcome lockdown. The lower "Negative" path describes serological assays (IgG/IgM) performed on asymptomatic workers. In case of a negative result, after 10 days of preventive quarantine, a second negative IgG/IgM serological test is required to return to work. In case of a positive result, a real-time PCR assay on a nasopharyngeal swab is performed. A negative result must be confirmed by a second real-time PCR assay performed 24 h apart before returning to work. The upper "Positive" path describes a positive real-time PCR assay performed on an individual who is IgG/IgM positive for SARS-CoV-2. The subject is quarantined for 14 days, and at the end of this period, a real-time PCR assay is performed on a new nasopharyngeal swab. A negative result must be confirmed by a second PCR assay carried out 24 h apart before the individual can return to work. Serological testing should be done with assays using coated spike protein or its subcomponents such as S1, S2 or RBD, which may elicit neutralizing antibodies against SARS-CoV-2.

use to identify people who can return to work safely as soon as the lockdown eases. Indeed, it is possible that we may have to cohabit with the virus for a significant period of time, and it is mandatory to develop strategies for reopening economic activities. A possible algorithm is outlined in Figure 1. According to this algorithm, a serological IgM/IgG assay should be performed on asymptomatic workers before returning to work. In the case of a negative result, the serological assay is repeated after 10 days and if confirmed negative, the subject returns to work. In case of positivity, a real-time PCR assay on a nasopharyngeal swab is performed. If positive, the subject is quarantined for 14 days, and at the end of quarantine, a real-time PCR assay is performed on a new nasopharyngeal swab. A negative result must be confirmed by a second PCR carried out 24 h apart before the individual can return to work.

However, considering the limitations of the current serological tests and the possibility of cross-reaction with SARS-CoV (that shares 82% nucleotide identity with SARS-CoV-2 [69]) and other coronaviruses, real-time PCR remains, to date, the main and most effective diagnostic test for COVID-19 worldwide.

5.1. Rapid immunochromatographic tests

Reliable rapid tests could alleviate the pressure on molecular biology laboratories where SARS-CoV-2 RNA is tested and could help in developing countries. In the European Union, licensed rapid tests include both qualitative and semi-quantitative tests [70].

There are two types of rapid tests: the SARS-CoV-2 antigen detection tests and antibody detection tests. To date, 10 CE-marked rapid SARS-CoV-2 antigen detection tests conform to the EU legislation FIND (https://www.finddx.org/), Directive 98/79/EC on IVDs. The sensitivity of these assays is generally low. For instance, the COVID-19 Ag Respi-Strip (Coris BioConcept, Brussels, Belgium), has a sensitivity of ~60% with 100% specificity (95% CI: 93.5-100%). Its positive predictive value is 100% (95% CI: 86.7–1.00%) and negative predictive value, 85.4% (95% CI: 75.4–91.9%). Agreement with real-time PCR is 88%.

As of 20 May 2020, there are over 60 rapid antibody detection tests. These tests have limited usefulness in the early diagnosis of COVID-19 disease because it may take 7–10 days after onset of symptoms for patients to become positive [62,66].

Rapid tests need validation on large number of samples before they can be introduced for diagnosis of SARS-CoV-2, and a comparison with CLIA or ELISA assays should be required. WHO referral laboratories are conducting validation studies on commercial assays [70]. These assays can be run on venous whole blood, fingerstick whole blood, serum and plasma. The sensitivities, specificities and accuracy vary with to the manufacturer.

As an example, the sensitivity of BasePoint™ COVID-19 IgG/IgM rapid Test Device (Abbott, Chicago, USA) is 86.43% (95% CI: 82.51–89.58%); specificity, 99.57% (95% CI: 97.63–99.92%); and accuracy, 91.61% (95% CI: 89.10–93.58%). In the case of the COVID-19 IgG/IgM Rapid Test (PRIMA Lab SA, Balerna, Switzerland), the accuracy of the test for IgG is 98.6% (specificity 98.0%, sensitivity 100.0%) whereas the accuracy for IgM is 92.9% (specificity 96.0%, sensitivity 85.0%). The 2019-nCoV IgG/IgM Rapid test Cassette (All Test Biotech Co., LTD, Hangzhou, China) states a sensitivity for IgM detection of 85.0% (95% CI: 82.1–96.8%), specificity of 96.0% (95% CI: 86.3–99.5%), and accuracy of 92.9% (95% CI: 84.1–97.6%). On the other hand, the sensitivity, specificity and accuracy for IgG detection is 100% (95% CI: 86.0–100%), 98.0% (95% CI: 89.4–99.9%), and 98.6% (95% CI: 92.3–99.96%) respectively. In any case, clinical validation in the field, performed by laboratory professionals, is strongly recommended for these rapid tests before their introduction in outpatient clinics or their use as direct-to-consumer testing. Recently, we compared rapid tests and a CLIA assay: the sensitivity for IgG was ~90% for the immunochromatographic tests and 95% for CLIA, while the sensitivity for IgM ranged from 61.4% to 87.8% for the immunochromatographic tests and was 91% for CLIA; the specificity was 100% for all [71].

Colloidal gold-based lateral flow immunoassays have been developed to detect antibody response against SARS-CoV-2 infection. These assays are typically qualitative (positive or negative), portable, easy to use, and rapid, and can be used at the point of care [72]. In a recent study, a colloidal gold lateral flow immunoassay was developed to detect IgM for SARS-CoV-2. The assay showed high sensitivity (100%) and specificity (93.3%) when compared to a real-time PCR assay performed on the serum of COVID-19 patients and healthy individuals, and almost a perfect agreement by K statistics (k coefficient = 0.872) [73]. In another study, a test for detecting IgM/IgG antibodies showed a sensitivity of 71.1% [95% CI 60.9–0.79.7%] and a specificity of 96.2% [95% CI 85.9–99.3%] [74]. Considering these technical features along with the ease of testing, low cost and delivery of results in a short time frame (10–15 min), colloidal gold-based lateral flow immunoassay, if properly validated, may be a suitable method for serologic screening in the context where a large number of samples have to be tested in a short time or where whole blood sampling is not possible.

6. Sars-CoV-2 and cellular infection

6.1. SARS-CoV-2 receptor

Several biochemical and structural studies have shown that SARS-CoV-2 binds the human receptor for angiotensin-converting-enzyme 2 (ACE2) [75–77]. The spike protein of SARS-CoV-2 contains a polybasic furin cleavage site at the boundary between subunit S1 and S2 of the spike protein that is processed during biogenesis and that distinguishes this virus from SARS-CoV and SARS-related coronavirus. This cleavage site is important for virus infectivity and host range. Six amino acids within the RBD of the spike protein are critical for binding to the ACE2 receptor and for determining the host range of SARS-like coronaviruses. Five of these six amino acids differ between SARS-CoV and SARS-CoV-2.

Structural studies carried out on the SARS-CoV-2 RBD/ACE2 complex showed that SARS-CoV-2 RBD binds ACE2 with higher affinity than SARS-CoV RBD. Indeed, SARS-CoV-2 RBD forms a larger binding interface and more contacts with ACE2 than SARS-CoV RBD [78]. From a structural point of view, there is a significant difference in the conformation of the loops in the ACE2 binding ridge that may explain this difference in receptor binding affinity between the two viruses. The SARS-CoV loop contains a three-residue motif, proline-proline-alanine, while SARS-CoV-2 and bat coronavirus RaGT13 contain a four-residue motif, glycine-valine/glutamine-glutamate/threonine-glycine. Due to these structural differences, the ridge in the SARS-CoV-2 RBD makes more contacts with the N-terminal helix of ACE2. The importance of this structure in determining the higher binding affinity of SARS-CoV-2 RBD for the ACE2 receptor was confirmed by mutations studies. Mutations at position 481–487, 493 and 501 of SARS-CoV-2, where the amino acids were mutated to those in SARS-CoV, reduced the surface of the SARS-CoV-2 spike available for binding to ACE2 [78]. The authors also showed that bat RaTG13 is able to bind ACE2. RaTG13 contains a four-residue motif in the ACE2 binding-ridge that is similar to SARS-CoV-2, suggesting that SARS-CoV-2 may have evolved from this virus or a bat-related coronavirus. Amino acid changes L486F and Y493Q from RaTG13 to SARS-CoV-2 facilitate ACE2 binding and support the idea that these changes enabled bat to human transmission. Also, L455 and N501, present both in RaTG13 and SARS-CoV-2, contribute to ACE2 binding and perhaps to bat to human transmission [78].

The intermediate host of SARS-CoV-2 is unknown. Pangolin has been proposed as the intermediate host for coronavirus transmission between bat and humans. The RBD of CoV-pangolin isolated in Guandong shows that it has several amino acids that favor binding to the human ACE2 receptor, supporting the hypothesis of pangolin as the intermediate host [78].

Taken together, these observations show that the SARS-CoV-2 RBD may be a useful target for antiviral drugs. By blocking the RBD region, the virus could be prevented from binding to the ACE2 receptor and entering the cell. Another possible target is TMPRSS2, a cellular serine protease used by SARS-CoV-2 for S priming and cell entry. Inhibition of TMPRSS2 by *Camostat mesylate* blocks SARS-CoV-2 infection in human hair cells [79].

6.2. Induction of T-cell lymphocytes apoptosis during human coronavirus infection

SARS-CoV-2 can infect T-cells lymphocytes *via* S protein mediated endocytosis and internalization [80], although the virus is not able to replicate in these cells. Interestingly, the translation of viral RNA proteins induces apoptosis in T-lymphocytes, as demonstrated for MERS coronavirus [81]. Induction of apoptosis may represent a mechanism of immune-evasion that contributes to the immune-pathogenicity of the virus as a result of the lymphocytopenia observed in COVID-19 patients.

Apoptosis, or programed cell death, is characterized by the controlled disassembly of cellular structures that are released as apoptotic bodies; these apoptotic bodies are then engulfed in the membranes of neighboring cells or by phagocytes [82]. Because the released material is surrounded by cellular membrane, this process is not inflammatory or immunogenic, in contrast to the process of necrosis in which the uncontrolled release of cytoplasmic cellular contents activates an inflammatory response. Fundamentally apoptosis can be activated by two pathways, the intrinsic pathway in which the cells receive an intrinsic death stimulus (oncogene activation, DNA damage or others), and the extrinsic pathway in which the death stimulus can be due to external factors such as Fas or TNF-α receptor activation during immuneresponse. Both pathways converge on the mitochondria, specifically to the outer membrane, and are controlled by the Bcl2 family proteins [82–84].

The apoptosis process has been studied in human coronavirus (HCoV) infected cells from SARS-CoV infected patients. Apoptotic cells have been detected in other tissues such as thyroid tissues [85], spleen and kidney [86], in addition to lung and upper airway epithelial cells. Apoptosis by human coronaviruses (HCoVs), including SARS-CoV, has been also described in *in vitro* systems and animals [87,88]. Recently, apoptosis was demonstrated in the spleen and lymph nodes of SARS-CoV-2 patients [89].

SARS-CoV infection is able to induce caspase dependent apoptosis, but even if this phenomenon is generated by the infection, apoptosis does not inhibit virus replication [90]. Apoptosis has been demonstrated in an *in vitro* experiment using cultured cell by transfecting single SARS-CoV coding regions, including S, E, M, N, and accessory protein 3a, 3b, 6, 7a, 8a, and 9b. The membrane protein, M, modulates the Akt survival pathway and release of mitochondrial cytochrome c [91], protein 3b activates bcl-2-like protein 4 (BAX), and both processes are able to induce apoptosis [92]. Even if the ablation of gene 7 from the SARS-CoV genome did not show an effect on replication of the virus in transformed cell lines, it showed a reduction in apoptosis if DNA fragmentation was evaluated [93]. The induction of apoptosis by protein 7a is blocked by Bcl-XL [94]. The proteins, E and 7a, are able to activate the intrinsic pathway by sequestering antiapoptotic Bcl-XL to the endoplasmic reticulum [87]. Protein 3 b is able to induce cell G0/G1 arrest and apoptosis [95]. Protein 3a is also pro-apoptotic, and possesses three major protein signatures, the cysteine-rich, Yxxϕ and diacidic domains, that are essential for its apoptotic function [96]. Other mechanisms of apoptosis induction by HCoV involve the activation of endoplasmic reticulum stress response and the mitogen-activated protein kinase pathway [97].

7. Safety issues

7.1. General safety recommendations

Since the outbreak of SARS-CoV-2, the use of face masks has become ubiquitous. The fear of being infected has caused everyone who can wear face mask to do so and this has contributed to the shortage of this product. Policies on wearing face masks differ among countries. WHO discourages the use of face-mask among healthy people unless they are taking care of a person with suspected SARS-CoV-2 infection or with respiratory symptoms. However, the use of face mask is always recommended because it could prevent infection transmission from asymptomatic carriers. In China, national policy encourages the use of face masks among people with low or moderate risk of infection but discourages those with a very low risk of infection from wearing a mask. People in quarantine should wear a face mask if they leave their home for any reason to prevent potential transmission in the asymptomatic phase. In addition, vulnerable populations such as the elderly or those with underlying medical conditions should wear a mask [98].

A recent study by Leung *et al.* showed that wearing face mask significantly reduced the shedding of respiratory viruses such as influenza virus and coronavirus [28]. Based on these recent findings and in an attempt to reduce the spread of SARS-CoV-2 in the so-called second phase of the epidemic, many EU governments have made it mandatory to wear face masks in public.

The potential air propagation of SARS-CoV-2 depends on many factors including the particle size, the speed of exhaled air (increased by breathing $<$ speaking $<$ coughing $<$ sneezing) as well as temperature and humidity.

The WHO, CDC and European Center for Disease Prevention and Control strongly recommend that people perform hand hygiene frequently and avoid touching their eyes, nose and mouth.

7.2. Blood specimens handling

Routine blood testing should be performed on automated analyzers using standard practices and procedures at containment level 2 for suspected/confirmed cases of COVID-19. Automated analyzers should be disinfected following sample processing, and actions that could generate infectious aerosols should be avoided. The opening of test tubes is usually not considered a high-risk procedure for generating aerosols. However, risk assessment should be undertaken to determine whether a microbiological safety cabinet should be used.

Blood samples from patients with COVID-19 disease are usually negative for SARS-CoV-2 RNA. Thus, viremia does not seem to be a major problem for SARS-CoV-2 infection. The virus has been detected only in a limited number of cases [11,31,99–102]. Furthermore, there is no clear correlation between the presence of viral RNA in the blood and the severity of disease [11,100,101]. The infectiousness of blood from patients with COVID-19 as well as SARS and MERS has not been demonstrated. No reports that show transmission of these three viruses by blood or blood products exist in the literature.

Based on this evidence, blood samples can be handled using standard laboratory practice at containment level 2. As a precaution, in case of severe COVID-19 disease, if laboratory procedures may result in splashes, spills or aerosol generation, a microbiological safety cabinet can be used to protect the operator [https://www.gov.uk/government/publications/wuhan-novel-coronavirus-guidance-for-clinical-diagnostic-laboratories/wuhan-novel-coronavirus-handling-and-processing-of-laboratory-specimens].

7.3. Respiratory samples

Respiratory samples may contain SARS-CoV-2, and therefore they must be processed in a microbiological safety cabinet at containment level 2; this includes preparation of specimens for molecular testing prior to sample inactivation, aliquoting or dilution of respiratory samples, and rapid antigen testing of respiratory specimens. Propagation or culturing of SARS-CoV-2 for diagnostic or research purposes must be conducted at containment level 3 [https://www.gov.uk/government/publications/wuhan-novel-coronavirus-guidance-for-clinical-diagnostic-laboratories/wuhan-novel-coronavirus-handling-and-processing-of-laboratory-specimens].

7.4. Surfaces disinfection

SARS-CoV-2 may persist on surfaces for a time ranging from hours to days [103]. Therefore, surfaces may represent a source of infection transmission. To minimize this risk, it is important to use disinfectants and detergents to kill SARS-COV-2. Several disinfectants are available on the market and can be used to clean surfaces that may be contaminated by the viruses: alcohols, peroxide and peroxy acids, quaternary ammonium compounds and inorganic compounds (chlorine, hypochlorite, hypochlorous acid) [104].

Recently the IFCC Task Force on Covid-19 published a set of recommendations, adapted from official documents of international and national health agencies, on biosafety measures for routine clinical chemistry laboratories [105].

8. Treatment

8.1. Determine the treatment site according to the condition of the patient

According to the severity of a patient's symptoms and the medical resources available in a region, different treatment sites may be selected to observe and isolate patients. The specific classification, from Chinese guidelines, is as follows:

A. Asymptomatic cases: They have not been confirmed and should not be considered as new cases. The main treatment measure is centralized quarantine for 14 days and further monitoring by the local Public Health Department [13]. If these cases are in home isolation, household members should stay in a different room, or if this is not possible, maintain a distance for at least 1 meter from the quarantined person [106];

B. Suspected cases: After informed consent, patients who have the ability to self-care, age $\leq$65 years old, without primary diseases such as respiratory diseases, cardiovascular diseases and mental health issues, should go to a health care facility voluntarily [107,108]. During quarantine observation, the person should in principle stay in a single room and not leave the room at random;

C. Mild cases: They are treated in a mobile cabin hospital if available or at home if hospitalization is not possible because of the heavy burden on the health care system. In this case, they should be followed up and cared for by family members [13]. If patients are in the same room, the space between beds should not be less than 1.2 m, and the room should be equipped with its own facilities. At the same time, family visits and nursing should be declined [108];

D. Severe/critically ill cases: Patients who are initially diagnosed as critically ill should be admitted into the Intensive Care Unit (ICU) immediately for treatment. For patients whose status changes from mild to severe, after hospital triage and prescreening expert consultation in the mobile cabin hospital or at home, they should be transferred to the critical observation and treatment area of a sheltered hospital, and following consultation, they should be transferred to a designated hospital for treatment [109].

Facing the COVID-19 pandemic, China creatively adopted large-scale mobile cabin hospitals to prevent and control the epidemic and achieved satisfactory results. Since the epidemic rapidly spread to many other countries around the world, China's experience is recommended and should be considered in their national strategy and at the local level. Thus, countries such as Serbia, Britain, Italy, Spain, France, Iran accelerated the construction of mobile cabin hospitals, which are used mainly to treat mild or asymptomatic patients. According to WHO guidelines, if patient symptoms are mild, providing care at home may be considered as long as patients can be followed up and cared for by family members [106]. However, severe cases should be admitted to designated hospitals for treatment at first diagnosis.

8.2. Treatment for mild COVID-19 cases

The clinical symptoms of these cases are mild and there is no pneumonia manifested on chest imaging. Patients should stay in bed, which is the principle for treatment of mild COVID-19 cases. According to the protocol by Prof. Tingbo Liang, blood oxygen saturation monitoring and oxygen therapy with nasal cannula should be conducted regularly for these patients.

8.3. Treatment for moderate COVID-19 cases

The clinical symptoms are moderate, with fever, respiratory tract symptoms and pneumonia manifestations on chest imaging. Treatment principles include bed rest, supportive treatment to maintain energy supply, maintaining water and electrolyte balance and monitoring vital signs and oxygen saturations.

Patients should be given oxygen therapy as needed. Firstly, they can be provided with nasal cannula therapy. If this does not work, patients should be treated with mask oxygen and cannula oxygen therapy. As well, inhalation of hydrogen-oxygen (O_2/H_2: 33.3%/66.6%) can be used.

Patients should also be given antiviral therapy. Lopinavir/ritonavir is an approved antiviral drug that blocks the cleavage of Gag-Pol polyprotein. Lopinavir is used to treat human immunodeficiency virus-1 infection, and its application in COVID-19 infection is mainly based on the treatment of MERS-CoV. Ribavirin is a synthetic nucleoside antiviral agent with broad-spectrum antiviral activity that can inhibit DNA and RNA viruses [110]. For the new coronavirus pneumonia, the latest Chinese clinical guidance recommends the combined application of ribavirin and interferon or lopinavir/ritonavir. Redoxivir has shown significant effect in the treatment of SARS and MERS virus infections. Holshue et al. treated the first case of COVID-19 infection in the United States with remdesivir, and the clinical symptoms and signs improved significantly after that [30]. In addition, antiviral drugs such as darunavir, abidor and fapiravir can theoretically be used in the therapy of COVID-19 infection, but their specific efficacy still needs to be verified by animal and clinical experiments.

A randomized double-blind, placebo-controlled, multicenter trial at ten hospitals in Hubei, China, was performed with remdesivir. The treatment was not associated with statistically significant clinical benefits. However, the duration of invasive mechanical ventilation, although also not significantly different between groups, was numerically shorter in remdesivir than placebo recipients [111].

During the treatment of MERS, α-interferon, a biologic drug, played an important role because of its antiviral activity [112]. The chinese clinical guidance recommends α-interferon for the treatment of COVID-19. Use of three or more antiviral drugs at the same time is not recommended, and blind or inappropriate use of antibacterial drugs (especially broad-spectrum antibacterial drugs) should be avoided [113].

8.4. Treatment for severe/critical COVID-19 cases

Severe cases have severe respiratory symptoms such as shortness of breath, a decrease of oxygen levels and a decrease of PaO_2/FiO_2. The general treatment principles for these cases are active prevention and treatment of complications, prevention of secondary infections while treating basic diseases, and organ function support treatment in a timely manner [114,115].

Detection of nucleic acid should be performed to monitor the replication of the virus (43). A series of indices should be tested including blood counts, plasma biochemical profile, routine urinalysis, stool, coagulation function, blood gas analysis, ASO, RF, CPR, cyclic citrullinated peptide, ESR, PCT, blood type, cardiac enzymes, respiratory virus tests, and cytokines [108]. The usefulness of serum (1, 3)-β-D-glucan and galactomannan assay in a suspected case of invasive pulmonary aspergillosis infection in a critically ill COVID-19 patient has been reported [116].

An ultrasound of the liver, gallbladder, pancreas and spleen should be done, and an echocardiogram and lung CT scan should be performed [108,113].

If the respiratory distress and/or hypoxemia cannot be relieved, high-flow nasal cannula oxygen therapy or noninvasive ventilation should be used. If the condition of the patient does not improve or if it worsens within 1–2 h, tracheal intubation and invasive mechanical ventilation should be performed. Sedation and muscle relaxants should be used for patients who have a problem with man-machine synchronization. Closed sputum suction should be considered according to airway secretions. For patients with severe acute respiratory distress syndrome (ARDS), lung expansion is recommended, and prone ventilation should be performed for more than 12 h every day. If it is not effective, extracorporeal membrane pulmonary oxygenation (ECMO) should be considered. The indications of ECMO are as follows: (1) when $FiO_2 > 90\%$, the oxygenation index is less than 80 mmHg for 3–4 h; (2) patients with simple respiratory failure with plateau ≥ 35 cm H_2O, the VV-ECMO mode is selected; if circulatory support is needed at the same time, the VA-ECMO mode is preferred. Weaning off ECMO can be considered when the underlying disease is under control and cardiopulmonary function shows signs of recovery [116,117].

For severe and critical cases, circulatory support therapy, renal function replacement therapy and blood purification therapy should be performed. For circulatory support therapy, improvement of the microcirculation and the use of vasoactive drugs can be considered based on adequate fluid resuscitation; attention should be paid to changes in the patient's heart rate, blood pressure, urine volume, arterial blood gas levels, and fluid balance. If the heart rate or blood pressure changes by more than 20% from baseline values, attention should be paid to septic shock, severe heart failure, or gastrointestinal bleeding [118]. Regarding renal function, the causes of renal function insufficiency (such as drugs and hypoperfusion) should be assessed, and treatment should focus on fluid balance, electrolyte balance and acid-base balance. Continuous renal replacement therapy (CRRT) can be used in severe cases to treat hyperkalemia, acidosis, pulmonary edema or excessive water load, and fluid management when multiple organ dysfunction occurs [119]. As the disease progresses, the patient will produce large amounts of inflammatory factors. The use of blood purification systems (such as plasma exchange, blood filtration, etc.) can remove inflammatory factors and reduce their damage to the patient. The indications for blood purification treatment for patients with severe COVID-19 can be divided into renal and non-renal. Blood purification treatment needs to be considered when patients with severe COVID-19 have acute kidney injury, (particularly stage ≥ 2 that meets the criteria of kidney disease), severe fluid overload, and electrolyte and acid-base balance disorders. Hemodialysis patients with hemodynamic instability should to change to CRRT. Non-renal indications include patients with severe ARDS, acute liver failure, multiple organ dysfunction syndrome, or septic shock, excessive inflammatory response and uncontrollably high fever (rectal temperature $>39.5\,^{\circ}C$) [120].

In addition, for the severe and critical cases, immunotherapy can be performed. For patients with extensive lung lesions and persistently elevated IL-6, monoclonal antibody therapy can be tried, but attention should be paid to allergic reactions. Immunotherapy is not recommended for patients with active infections, e.g. tuberculosis [116].

For patients with deterioration of oxygenation, rapid imaging changes and excessive inflammatory response, glucocorticoids can be used as appropriate. Experts from the Chinese Thoracic Society developed a consensus statement that patients can be treated with glucocorticoids when they meet the following four conditions at the same time: (1) adult (age $\geq$18 years old); (2) diagnosed with new coronavirus infection by PCR or serum antibodies; (3) symptoms (including fever, cough or other related infection symptoms) occur within 10 days, radiographically confirmed pneumonia and rapid progress; and (4) blood oxygen saturation (SpO2) $\leq$93% or shortness of breath (breath rate $\geq$30 breaths/min) or oxygenation index $\leq$300 mmHg [121]. For patients with secondary infection, intestinal microecological regulators and broad-spectrum antibacterial drugs can be used. Immunoglobulin infusion therapy can also be used to improve patient immunity [122]. It is recommended that pregnant women with severe COVID-19 infection end their pregnancy as soon as possible. In addition, anxiety and fear are often present in severe and critical cases, and the psychiatric service must have a specific approach for dealing with these problems including hospital resource management, mental health intervention, and guidance for family members [123].

8.5. Traditional Chinese medicine treatment

Traditional Chinese medicine (TCM) is a unique diagnostic method with a systematic approach, abundant historical literature and materials. It formulates treatment based on symptom-based diagnosis, an approach that is increasingly emphasized by modern medicine. In the treatment of COVID-19, TCM is proposed as an important option by national and provincial guidelines with substantial utilization. According to TCM theory, COVID-19 belongs to the category of "dampness plague", and its core pathogenesis is correlated to "epidemic virus." The "state" of cold and humidity is the basis of the disease, and "stagnation of qi" is the key to disease progression. The corresponding treatment at the early stage targets eliminating dampness, and the strategy focuses on eliminating dampness, releasing lungs and expelling pathogenic factors, to shorten fever duration, relieve symptoms, prevent disease progression, reduce mortality and assist rehabilitation. For TCM treatment, the 7th edition of China National Clinical Guideline divided COVID-19 into a medical observation period and treatment period. The latter is further stratified into four manifestations including mild, intermediate, severe, and critical, followed by a rehabilitation stage [124].

8.6. Prospects for COVID-19 drug research

At present, there is no specific drug for the treatment of COVID-19, and the main treatment methods are supportive and symptomatic treatment [125]. The mechanisms of different therapeutic drugs mainly include blocking virus invasion, inhibiting virus replication and protein expression and regulating the patient's immunity. Up to 21 April 2020, there were 159 drugs, including 49 antibody drugs, 20 antiviral drugs, 12 cell-based therapy drugs, 5 RNA therapy drugs, and others, in various stages of research [126]. In addition to the therapeutic drugs mentioned above, the antimalarial drugs, chloroquine and hydroxychloroquine, are also considered as candidates for the treatment of patients with COVID-19. Both drugs exert an inhibitory effect on the virus by interfering with the binding of ACE2 [127]. Chloroquine phosphate drug treatment has been included in clinical trials, and a phase III clinical trial for the drug was started in the United States on 31 March 2020 [128].

A potential promising treatment target is cathepsin L (CatL), an endosomal cysteine protease that mediates the cleavage of the S1 subunit of the coronavirus surface spike glycoprotein. While TMPRSS2 acts locally at the host cell plasma membrane and possibly during endocytotic vesicle trafficking, CatL continues S1 subunit degradation in the acidic endosome and lysosome compartments. Therefore, CatL inhibition could provide two sequential blocks for coronavirus infection: blockage of virus entry on the host cell surface and blockage of viral material release and replication inside the host cell endosomes [129].

Finally, plasma provided by COVID-19 convalescent patients may be used as therapy in critically ill patients infected by SARS-CoV-2. Convalescent plasma therapy played a role in the treatment of SARS [130]. Clinical trials have shown that after receiving convalescent plasma therapy, the patient's inflammation index and viral load decreased significantly, and blood oxygen saturation improved. A strict protocol to enroll convalescent volunteers and to assess the level of immunogenicity of plasma is needed [131].

We believe that with the joint efforts of various countries on the research and development of COVID-19 drugs, we will soon find the most effective drugs for its therapy.

9. Prevention and vaccination

According to the infection prevention and control strategies from the WHO, standard precautions for all patients, which are also appropriate for public prevention, include hand and respiratory hygiene, the use of appropriate personal protective equipment, safe injection practices, safe waste management, clean linens, environmental cleaning, and sterilization of patient-care equipment [132]. A vaccine to prevent COVID-19 is perhaps the best hope for ending the pandemic, and vaccines are especially needed by health care workers on the front lines and other vulnerable members of the population who have a higher risk of contracting the infection.

Currently, researchers are racing to develop such a vaccine. With international alliances and government efforts to organize resources to make multiple vaccines urgently on shortened timelines, 115 vaccine candidates are in development, with several vaccines in clinical trials by now [133]. According to China's National Health Commission, there are currently five technological approaches to develop COVID-19 vaccines in China, namely inactivated vaccines, genetic engineering subunit vaccines, adenovirus vector vaccines, nucleic acid vaccines, and vaccines using attenuated influenza virus as vectors [134]. Most teams completed preclinical research in April 2020.

Coronaviruses have a spike-like structure, the S protein, on their surface. The S protein is considered to be the key molecule that binds to the ACE2 receptor to attach onto the surface of human cells and that forces the virus through the cell membrane, mediating the process of virus infection and membrane fusion to host cells [135]. With the exception of inactivated vaccines and attenuated virus vaccines, other vaccines are designed based on the S protein. The S protein could be expressed *in vivo* using vectors carrying the S gene, and the expressed protein would induce the synthesis of specific antibodies that could block the entry of the virus to the cells. These approaches are expected to prevent the virus from binding to human cells and to stop the virus from reproducing so as to achieve immune protection. The most advanced candidates that have moved into clinical development are shown in Table 2 [133,136].

While injection is the most widely used way for vaccine delivery, there are also some novel approaches to deliver vaccine. For example, a fingertip-sized microneedle array vaccine, PittCoVacc, a vaccine from University of Pittsburgh School of Medicine, delivers the spike protein pieces into the skin to stimulate the body to produce antibodies against SARS-CoV-2 [137]; INO-4800 delivers DNA into the host's cells by a handheld smart device, CELLECTRA [138]; and Symvivo COVID-19 products are presented as oral lyophilized gel-capsules [139].

Different types of technological platforms have their own characteristics regarding speed of research and development, production cycle, equipment requirements, etc. The safety, stability and effectiveness of vaccine candidates still need to be evaluated and compared through clinical experiments. Advantages and disadvantages of different COVID-19 vaccine technology strategies are listed in Table 3 [140].

During the global emergency of the COVID-19 pandemic, the effort in response is unprecedented in terms of scale and speed. However, much work remains unfinished. The vaccine candidates might prove to be not safe or effective in any phase of development; for instance, the Coalition for Epidemic Preparedness Innovations indicates a potential success rate of only 10% for vaccine development [133].

Recently a study about the safety, tolerability and immunogenicity of a recombinant adenovirus type-5 vectored COVID-19 vaccine was performed in Wuhan; neutralizing antibodies increased significantly at day 14 and peaked 28 days post-vaccination whereas the T-cell response peaked at day 14 post-vaccination [141].

Strong international coordination and cooperation among research groups involved in vaccine research and development will be needed to ensure that promising vaccines can be manufactured in sufficient quantities and equitably supplied to all affected areas, particularly low-resource regions.

A prophylactic vaccine against SARS-CoV-2 is particularly urgent because SARS-CoV-2 is a novel virus of zoonotic origin [1,2] that started spreading among human beings, probably in late November 2019 [11], to which the population does not have an existing level of immunity that is able to limit or stop the circulation of the virus. Assuming a R_0 of 3 for SARS-CoV-2, the herd immunity threshold for SARS-CoV-2 is about 67% [142]. This means the infection will start declining once the number of individuals with acquired immunity to SARS-CoV-2 exceeds 0.67 [142]. Acquisition of herd immunity *via* natural infection is ethically unacceptable because it would cause millions of deaths around the globe. Furthermore, the unchecked spread of SARS-CoV-2 will quickly paralyze our health care systems, increasing not only the COVID-19 mortality but also all causes mortality. In light of this, and because the development of an effective vaccine will take at least 12–18 months, it will be important to adhere strictly to preventive measures indicated by the WHO and to maintain social distancing.

Another important aspect that merits consideration is the duration of protective immunity. To date, we do not know how long protective immunity will last. Previous studies have demonstrated that neutralizing

Table 2. Clinical-phase vaccine candidates for COVID-19.

Vaccine candidate	Technology	Vaccine characteristics	Trail status	Developer	Location	References
Inactivated Novel Coronavirus Pneumonia vaccine (Vero cells)	Inactivated virus	Inactivated SARS-CoV-2 virus (Vero cells)	Phase I(ChiCTR2000031809)	Beijing Institute of Biological Products, Wuhan Institute of Biological Products	China	-Wuhan Institute of Virology, Wuhan Institute of Biological Products co., LTD. A randomized, double-blind, placebo parallel-controlled phase I/II clinical trial for inactivated Novel Coronavirus Pneumonia vaccine (Vero cells) (Registration number: ChiCTR2000031809) [EB/OL]. (2020-04-13) [2020/04/29]. http://www.chictr.org.cn/showprojen.aspx?proj=52227. - Tuna Y I. Wuhan Institute of Biological Products' COVID-19 Vaccine Enters Phase II Trials[J]. EqualOcean, 2020.
PiCoVacc	Inactivated virus	Inactivated SARS-CoV-2 virus	Phase I-II(NCT04352608)	Sinovac Biotech Ltd, Institute of Laboratory Animal Science, China	China	- Sinovac Biotech Co., Ltd. Safety and Immunogenicity Study of Inactivated Vaccine for Prophylaxis of SARS CoV-2 Infection (COVID-19) (ClinicalTrials.gov Identifier: NCT04352608) [EB/OL]. (2020-04-28) [2020/04/29] - Gao Qiang, Bao Linlin, Mao Haiyan, et al. Rapid development of an inactivated vaccine candidate for SARS-CoV-2[J]. Science, 2020:c1932. DOI:10.1126/science.abc1932.
Ad5-nCov	Non-replicating viral vector	Recombinant adenovirus type 5 vector	Phase I(NCT04313127); Phase II interventional trial for dosing and side effects (NCT04341389)	CanSino Biologics, Institute of Biotechnology of the Academy of Military Medical Sciences	China	- Institute of Biotechnology, Academy of Military Medical Sciences, PLA of China, CanSino Biologics Inc. A Phase II Clinical Trial to Evaluate the Recombinant Vaccine for COVID-19 (Adenovirus Vector) (CTII-nCoV) (ClinicalTrials.gov Identifier: NCT04341389) [EB/OL]. (2020-04-21) [2020/04/29/]. https://www.clinicaltrials.gov/ct2/show/NCT04341389. - Institute of Biotechnology, Academy of Military Medical Sciences, PLA of China, CanSino Biologics Inc. Phase I Clinical Trial of a COVID-19 Vaccine in 18-60 Healthy Adults (CTCOVID-19) (ClinicalTrials.gov Identifier: NCT04313127) [EB/OL]. (2020-04-14) [2020/04/29]. https://www.clinicaltrials.gov/ct2/show/NCT04313127. - Keown A. China's CanSino Prepares to Advance COVID-19 Vaccine Candidate into Phase II[J]. BioSpace, 2020.
ChAdOx1 nCoV-19	Non-replicating viral vector	Adenovirus vector	Phase I-II(NCT04324606)	University of Oxford	United Kingdom	- University of Oxford. A Study of a Candidate COVID-19 Vaccine (COV001) (ClinicalTrials.gov Identifier: NCT04324606) [EB/OL]. (2020-04-24) [2020/04/29]. https://www.clinicaltrials.gov/ct2/show/NCT04324606. - COVID-19 Oxford Vaccine Trial [EB/OL]. (2020-04-25) [2020/04/29]. https://www.covid19vaccinetrial.co.uk/about. - U.K. Starts Oxford Coronavirus Vaccine Trial as Germany Green-Lights BioNTech and Pfizer [EB/OL]. (2020-04-23) [2020/04/29/]. https://www.genengnews.com/news/uk-starts-oxford-coronavirus-vaccine-trial-as-germany-green-lights-clinical-trial-for-biontech-and-pfizer.
INO-4800	DNA	DNA plasmid encoding S protein delivered by electroporation (hand-held smart device "CELLECTRA")	Phase I-II(NCT04336410)	Inovio Pharmaceuticals, CEPI, Korea National Institute of Health, International Vaccine Institute	United States, South Korea	- Inovio Pharmaceuticals I. Inovio Accelerates Timeline for COVID-19 DNA Vaccine INO-4800[J]. PR Newswire, 2020. - Inovio Pharmaceuticals. Safety, Tolerability and Immunogenicity of INO-4800 for COVID-19 in Healthy Volunteers (ClinicalTrials.gov Identifier: NCT04336410) [EB/OL]. (2020-04-24)[2020/04/29]. https://www.clinicaltrials.gov/ct2/show/NCT04336410.

(*continued*)

Table 2. Continued.

Vaccine candidate	Technology	Vaccine characteristics	Trail status	Developer	Location	References
bacTRL-Spike	DNA	Bacterial medium with *Bifidobacterium longum* engineered to deliver DNA plasmid encoding S protein (oral)	Phase I(NCT04334980)	Symvivo Corporation, University of British Columbia, Dalhousie University	Canada	- Symvivo. COVID-19 Program Vision [EB/OL]. (2020-04-29) [2020/04/29/]. https://www.symvivo.com/covid-19. - Corporation S. Evaluating the Safety, Tolerability and Immunogenicity of bacTRL-Spike Vaccine for Prevention of COVID-19 (ClinicalTrials.gov Identifier: NCT04334980) [EB/OL]. (2020-04-22)[2020/04/29/]. https://clinicaltrials.gov/ct2/show/NCT04334980.
mRNA-1273	RNA	LNP-encapsulated mRNA vaccine encoding S protein	Phase I (NCT04283461)	Moderna, US National Institute of Allergy and Infectious Diseases	United States	- Moderna Therapeutics, US NIAID. Safety and Immunogenicity Study of 2019-nCoV Vaccine (mRNA-1273) for Prophylaxis SARS CoV-2 Infection (COVID-19) (ClinicalTrials.gov Identifier: NCT04283461) [EB/OL] (2020-04-13) [2020/04/29/]. https://www.clinicaltrials.gov/ct2/show/NCT04283461. - Grady D. Trial of Coronavirus Vaccine Made by Moderna Begins in Seattle[J]. New York Times, 2020.
BNT162 (a1, b1, b2, c2)	RNA	LNP-encapsulated mRNA vaccine encoding S protein	Phase I-II (UTRN: U1111-1249-4220)	BioNTech, Fosun Pharma, Pfizer	Germany, China, United States	- Pharmaceuticals B R G. A Multi-site Phase I/II, 2-Part, Dose-Escalation Trial Investigating the Safety and Immunogenicity of four Prophylactic SARS-CoV-2 RNA Vaccines Against COVID-2019 Using Different Dosing Regimens in Healthy Adults (UTRN: U1111-1249-4220) [EB/OL]. (2020-04-20) [2020/04/29]. https://www.clinicaltrialsregister.eu/ctr-search/trial/2020-001038-36/DE. - Reuters. Germany Approves Trials of COVID-19 Vaccine Candidate[J]. New York Times, 2020.
LV-SMENP-DC	Unknown	DCs modified with lentiviral vector expressing synthetic minigene based on domains of selected viral proteins; administered with antigen specific CTLs	Phase I(NCT04276896)	Shenzhen Geno-Immune Medical Institute	China	Shenzhen Geno-Immune Medical Institute. Immunity and Safety of Covid-19 Synthetic Minigene Vaccine (ClinicalTrials.gov Identifier: NCT04276896) [EB/OL]. (2020-03-19)[2020/04/29]. https://www.clinicaltrials.gov/ct2/show/NCT04276896.
Covid-19/aAPC	Unknown	aAPCs modified with lentiviral vector expressing synthetic minigene based on domains of selected viral proteins	Phase I(NCT04299724)	Shenzhen Geno-Immune Medical Institute	China	Shenzhen Geno-Immune Medical Institute. Safety and Immunity of Covid-19 aAPC Vaccine (ClinicalTrials.gov Identifier: NCT04299724) [EB/OL]. (2020-03-09)[2020/04/29]. https://www.clinicaltrials.gov/ct2/show/record/NCT04299724.

Table 3. Advantages and disadvantages of different COVID-19 vaccine technology strategies.

Technology	Characteristics	Advantages	Disadvantages
Inactivated vaccine	Consisting of virus particles with immunogenicity but reduced infectivity (virulence) which have been grown in culture and fully destroyed using heat, chemicals, or radiation.	• Similar antigenicity to live virus due to unchanged structure virus particles • No virulence rebound risk • Non-replicatable in host • No virus transmission risk among people • Relatively mature technology • Ease of preparation and production	• Multiple-dose and adjuvants may be required due to insufficient immunogenicity • Uncontrollable protective effect • BSL-3 laboratory required
Genetic engineering subunit vaccines	A subunit vaccine is a fragment of a pathogen, typically a surface protein, that is used to trigger an immune response and stimulate acquired immunity against the pathogen from which it is derived. For COVID-19, the subunit vaccine is the recombinant Spike protein or its peptide prepared in vitro through genetic engineering.	• Ability to induce robust transgene-specific T cell and antibody responses • Safety guaranteed because of no virus genome being injected	• Multiple-dose and adjuvants may be required
Adenovirus vector vaccines	Adenoviruses as vectors for delivering genes or vaccine antigens to the target host	• Broad range of tissue tropism • Well-characterized genome • Ease of genetic manipulation including acceptance of large transgene DNA insertions • Inherent adjuvant properties • Ability to induce robust transgene-specific T cell and antibody responses • Non-replicative nature in host • Ease of production at large scale	• Pre-existing immunity in humans, inflammatory responses • Sequestering of the vector to liver and spleen • Immunodominance of the vector genes over transgenes.
Nucleic acid vaccines	Genetically engineered plasmid containing the DNA sequence or RNA containing vector which could produce the Spike protein of SARS-COV-2 upon the delivery of the vaccine into the body.	• No risk for infection • Immune response focused on antigen of interest • Ease of development and production • Stability for storage and shipping • Cost-effectiveness • Obviates need for peptide synthesis, expression and purification of recombinant proteins and use of toxic adjuvants • Long-term persistence of immunogen • In vivo expression ensures protein more closely resembles normal eukaryotic structure, with accompanying post-translational modifications	• Limited to protein immunogens • Risk of affecting genes controlling cell growth • Possibility of inducing antibody production against DNA • Possibility of tolerance to the antigen (protein) produced • Unclear vaccine delivery efficiency and safety problem
Vaccines using attenuated influenza virus as vectors	Attenuated influenza virus as vectors for delivering the Spike protein of SARS-COV-2 to the target host	• Protect against COVID-19 and influenza at the same time • Convenient inoculation by nasal drip • The vector is little influence by the human immune system • No risk of integrating vector's DNA into the host genome	• Vector capacity limits the range of exogenous genes inserted • Less targeted
Live attenuated virus vaccine	Reducing the virulence of a pathogen, but still keeping it viable.	• Activates all phases of the immune system • Provides more durable immunity; boosters are required less frequently • Low cost • Some are easy to transport and administer (for instance orally) • Vaccines have strong beneficial nonspecific effects	• Possibilities that the virus can revert to wild type or develop into an entirely new strain • Safety problems.Potentially severe complications • Live strains typically require advanced maintenance, making transport difficult and costly • BSL-3 laboratory required

antibodies toward SARS-CoV lasted several months to two years, although low antibody titers were measured in all patients after about 15 months [143]. A decrease in antibody titer has also been reported for the 229E coronavirus that is responsible for the common cold [144], and this antibody titer was not sufficient to prevent reinfection. If this scenario applies to SARS-CoV-2, protective immunity will decrease over time and herd immunity will never be attained unless there is recurrent vaccination.

10. Artificial intelligence and mobile health tools

Artificial intelligence (AI) may have a role in the governance of health care systems and in coping with health emergencies such as the current COVID-19 outbreak. AI may help in analyzing a huge amount of data in a timely way as required during periods of crisis, allowing a prompt response by health authorities. Analysis of medical records, therapies, and laboratory findings may speed up using AI, hasten the decision-making process and improve patient management [145]. For instance, it has been proposed that AI could support radiologists in reading CT scans. While a manual read takes about 15 min, AI can complete the reading in a few seconds. Thus, AI could be designed to detect lesions resembling coronavirus pneumonia, to measure the density, volume and shape, and to compare multiple lung lesions from the image. This information should support physician in making a more rapid diagnosis [146]. In the work by Li and colleagues, AI allowed the detection of COVID-19 pneumonia and distinguished it from community acquired pneumonia and other lung lesions [147].

Block chain and AI could be used for remote patient monitoring and the transfer of clinical information to health authorities [145]. A positive SARS-CoV-2 patient could be referred to a quarantine site for monitoring and treatment to limit virus spread. Similarly, information from a certain geographic area could be used for tracking positive patients and/or quarantining that geographic area to limit the spread of the virus.

A mobile-phone based survey along with an AI framework has been proposed to collect travel history along with the common clinical manifestations to identify people with suspected SARS-CoV-2 infection. The collected information should stratify individuals under investigation into no-risk, minimal-risk, moderate-risk, and high-risk of being infected by the virus. Identification of the high-risk individuals should trigger immediate quarantine to contain the spread of the virus [148]. In Italy, the mobile-phone app, Immuni, was introduced and can be downloaded to trace contacts of an infected individual using Bluetooth technology. In case of positivity to SARS-CoV-2, the individual uploads the laboratory result on the platform, and an instant message is sent to all close contacts who must remain in quarantine for at least 14 days [www.salute.gov.it].

A pre-trained deep learning-based drug-target interaction model called Molecule Transformer-Drug Target Interaction (MT-DTI) was used to identify molecules already available on the market that could be used against SARS-CoV-2. Several molecules were identified including atazanavir, remdesivir, efavirenz, ritonavir, and dolutegravir. Atazanavir, the best compound, showed an inhibitory potency with a *Kd* of 94.94 nmol/L against the SARS-CoV-2 3C-like proteinase, followed by remdesivir (113.13 nmol/L), efavirenz (199.17 nmol/L) and ritonavir (204.05 nmol/L). Lopinavir and ritonavir target proteases, and based on this model could also inhibit the replication components of SARS-CoV-2 with a $Kd < 1000$ nmol/L [149].

DeepMInd developed an AlphaFold algorithm to predict protein structures starting from their amino acid sequences, thus avoiding long and intensive laboratory experiments. Using a deep neural network, the system can make accurate predictions of the distances and angles between amino acid residues and provide more information about the structure than contact predictions. The information obtained might be used as a platform for the development of therapeutics [150]. This algorithm works well even on sequences with fewer homologous sequences, as is the case of SARS-CoV-2.

11. Preparedness and lifestyle after COVID-19 pandemic

The COVID-19 pandemic has shown that even the most advanced health care systems cannot sustain a massive influx of critically ill patients in their emergency departments. Italy, with its 3.2 hospital beds per 1000 persons *vs* 2.8 in the United States had enormous difficulty to meet the needs of critically ill patients arriving in the hospitals in a short time frame [151]. As a consequence, elective or semi-elective surgical procedures were canceled or postponed, wards were reorganized to treat COVID-19 patients, and follow up visits were delayed.

To prevent this massive influx to emergency departments in a possible second wave, the implementation of regional assistance is crucial. A large number of nasopharyngeal swabs and serological testing should be performed in the region to identify infected people and to allow contact tracing. Then, patients with mild symptoms or asymptomatic carriers could stay at home and be monitored by health operators, thus reducing visits to clinics. Limiting access to hospitals will save resources, make them available to those who really need them and reduce the risk of exposure. In light of that, the Italian government has recently decided to increase the number of nurses in the region, bringing the ratio to 8 nurses/50,000 persons. Special units of continuing care (Italian, urban search and rescue) have been created to assist patients at home.

In addition, the number of beds in the ICUs will be increased by 70% and to a lesser extent, those in sub-ICUs. Finally, 600 beds will be made available for movable structures in the case of epidemic peaks. Meantime, it is necessary to increase the numbers of ventilators, ECMO machines, and personal protective equipment.

COVID-19 prevalence is currently determined by the number of subjects with positive RT-qPCR nasopharyngeal swabs, but the real prevalence is probably much higher. Testing for the presence of IgG anti-SARS-CoV-2 may identify those who have been exposed and asymptomatic carriers, giving us more reliable case counts and mortality estimates, and a useful tool to control the restarting phase.

Indeed, in a very short time, all countries will be challenged by phase two. However, until the effect of the neutralizing antibodies detected by all the different methods and the antibody serum levels that are needed for individuals to be fully protected against reinfection are known, we could not consider serological positivity as a "license" to stop social distancing rules and the use of protective devices.

Then, during this second phase, it will be important to maintain social distancing, which remains the major preventive measure in the absence of a vaccine and effective antiviral drugs. Social distancing implies a massive reorganization of our society and lifestyle.

Telecommuting (working at a distance) and smart working should be pursued whenever possible. When this is not possible, social distancing and protective measures (protective equipment, hand hygiene, disinfectants) must be provided in the workplace. Since large gatherings will not be permitted, commercial activities such as restaurants, cafes, cinemas, etc. will have to reduce the number of employees with consequent social repercussions and increased poverty. Welfare will be essential for such workers.

Schools and universities will be required to reorganize themselves to provide education and safety for their students. Access to the Internet will be crucial for all students and should be provided to enable them to pursue education and take exams remotely if necessary.

Finally, communication will be important to live through this difficult time. Clear messages from national authorities will be necessary to increase awareness in the population and to support them in following new norms.

Conclusions

The COVID-19 pandemic has stressed our health care systems in an unprecedented way and underlined once more the essential role of laboratory medicine in tackling the spread of new transmissible agents.

Almost all over the world, networks of COVID laboratories have been set up to support the specific needs of citizens and patients, and they will continue to be fundamental during the re-starting of social and work activities.

Traditionally laboratory medicine has been considered to be an integral part of the decision making process that supports 70% of clinical decisions [152]; in the COVID-19 era, this contribution may be even higher and close to 100% because decisions like patient admission, isolation and/or discharge are taken based on laboratory results [153].

The COVID-19 pandemic has revealed the weaknesses of our health systems that were unprepared to cope with a very large number of patients requiring respiratory support therapy in a short time frame. This emergency forced health authorities to stop all non-urgent medical procedures and convert wards in ICUs or sub-ICUs. On the other hand, the pandemic has prompted the scientific community to join together in efforts to fight this novel pathogen. Within a few days from the first reported cases of unknown pneumonia, the virus was isolated, sequenced, identified and genetically characterized. It was named SARS-CoV-2 because of its phylogenetic relationship with SARS-CoV and bat SARS-like coronaviruses. Based on its genetic features, molecular and serological assays were developed and have been introduced in routine diagnostics. Furthermore, several vaccine strategies have been developed or are in development, and trials are ongoing or will be begun to determine their effectiveness. In the absence of effective and specific antiviral therapy against SARS-CoV-2, the availability of a prophylactic vaccine is crucial.

Finally, a contribution in the governance of such an emergency could come from AI. Faster processing of clinical data may allow physicians to speed up the decision making process and improve the management of patients. AI and blockchain already used to trace contacts to contain the spread of the virus. In the near future, this technology will become an integral part of our health care systems.

Acknowledgements

Italy was the first country engaged in fighting SARS-CoV-2 pandemic after China. This is the reason we wrote this review with our Chinese colleagues. We thank them for their support that came from their experience, for their scientific contribution, and for their friendship.

Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

Sergio Bernardini http://orcid.org/0000-0003-1984-6834

References

[1] Zhou P, Yang XL, Wang XG, et al. A pneumonia outbreak associated with a new coronavirus of probable bat origin. Nature. 2020;579(7798):270–273.
[2] Wu F, Zhao S, Yu B, et al. A new coronavirus associated with human respiratory disease in China. Nature. 2020; 579(7798):265–269.
[3] World Health Organization Coronavirus Disease 2019 (COVID-19) Situation Report-97. Available from: https://www.who.int/docs/default-source/coronaviruse/situation-reports/20200426-sitrep-97-covid-19.pdf
[4] Wang C, Horby PW, Hayden FG, et al. A novel coronavirus outbreak of global health concern. Lancet. 2020;395(10223):470–473.
[5] Hui DSC, Zumla A. Severe acute respiratory syndrome: historical, epidemiologic, and clinical features. Infect Dis Clin North Am. 2019;33(4):869–889.
[6] Azhar EI, Hui DSC, Memish ZA, et al. The Middle East respiratory syndrome (MERS). Infect Dis Clin North Am. 2019;33(4):891–905.
[7] Corman VM, Muth D, Niemeyer D, et al. Hosts and sources of endemic human coronaviruses. Adv Virus Res. 2018;100:163–188.
[8] Andersen KG, Rambaut A, Lipkin WI, et al. The proximal origin of SARS-CoV-2. Nat Med. 2020;26: 450–452.
[9] Almazán F, Sola I, Zuñiga S, et al. Coronavirus reverse genetic systems: infectious clones and replicons. Virus Res. 2014;189:262–270.
[10] Nao N, Yamagishi J, Miyamoto H, et al. Genetic predisposition to acquire a polybasic cleavage site for highly pathogenic avian influenza virus hemagglutinin. mBio. 2017;8(1):e02298.
[11] Huang C, Wang Y, Li X, et al. Clinical features of patients infected with 2019 novel coronavirus in Wuhan, China. Lancet. 2020;395(10223):497–506.
[12] Wang D, Hu B, Hu C, et al. Clinical characteristics of 138 hospitalized patients with 2019 novel coronavirus-infected pneumonia in Wuhan, China. JAMA. 2020;323(11):1061.
[13] Zhu N ,Zhang D ,Wang W, et al. A novel coronavirus from patients with pneumonia in China, 2019. N Engl J Med. 2020;382(8):727–733.
[14] Chen N, Zhou M, Dong X, et al. Epidemiological and clinical characteristics of 99 cases of 2019 novel coronavirus pneumonia in Wuhan, China: a descriptive study. Lancet. 2020;395(10223):507–513.
[15] Lippi G, Plebani M. The critical role of laboratory medicine during coronavirus disease 2019 (COVID-19) and other viral outbreaks. Clin Chem Lab Med. 2020;58(7):1063–1069.
[16] Bhargava A, Fukushima EA, Levine M, et al. Predictors for severe COVID-19 infection. Clin Infect Dis. 2020. DOI:10.1093/cid/ciaa674
[17] Wang CZ, Hu SL, Wang L, et al. Early risk factors of the exacerbation of coronavirus disease 2019 pneumonia. J Med Virol. 2020. DOI:10.1002/jmv.26071
[18] Hamming I, Timens W, Bulthuis ML, et al. Tissue distribution of ACE2 protein, the functional receptor for SARS coronavirus. A first step in understanding SARS pathogenesis. J Pathol. 2004;203(2):631–637.
[19] Renu K, Prasanna PL, Valsala Gopalakrishnan A. Coronaviruses pathogenesis, comorbidities and multi-organ damage - a review. Life Sci. 2020;255: 117839.
[20] Long B, Brady WJ, Koyfman A, et al. Cardiovascular complications in COVID-19. Am J Emerg Med. 2020. DOI:10.1016/j.ajem.2020.04.048
[21] Ruan Q, Yang K, Wang W, et al. Clinical predictors of mortality due to COVID-19 based on an analysis of data of 150 patients from Wuhan, China. Intensive Care Med. 2020;46(5):846–848.
[22] Lippi G, Favaloro EJ. D-dimer is associated with severity of coronavirus disease 2019: a pooled analysis. Thromb Haemost. 2020;120(05):876–878.
[23] Lang J, Yang N, Deng J, et al. Inhibition of SARS pseudovirus cell entry by lactoferrin binding to heparan sulfate proteoglycans. Plos One. 2011;6(8): e23710.
[24] Vicenzi E, Canducci F, Pinna D, et al. Coronaviridae and SARS-associated coronavirus strain HSR1. Emerging Infect Dis. 2004;10(3):413–418.
[25] Belen-Apak FB, Sarialioglu F. The old but new: can unfractioned heparin and low molecular weight heparins inhibit proteolytic activation and cellular internalization of SARSCoV2 by inhibition of host cell proteases? Med Hypotheses. 2020;142:109743.
[26] Henry BM, Benoit SW, Santos de Oliveira MH, et al. Laboratory abnormalities in children with mild and severe coronavirus disease 2019 (COVID-19): a pooled analysis and review. Clin Biochem. 2020.;81: 1–8. DOI:10.1016/j.clinbiochem.2020.05.012
[27] Sanna G, Serrau G, Bassareo PP, et al. Children's heart and COVID-19: Up-to-date evidence in the form of a systematic review. Eur J Pediatr. 2020. DOI: 10.1007/s00431-020-03699-0

[28] Leung NHL, Chu DKW, Shiu EYC, et al. Respiratory virus shedding in exhaled breath and efficacy of face masks. Nature Med. 2020;26(5):676–680.

[29] Han Q, Lin Q, Ni Z, et al. Uncertainties about the transmission routes of 2019 novel coronavirus. Influenza Other Respir Viruses. 2020;14(4):470–471.

[30] Zhao S, Lin Q, Ran J, et al. Preliminary estimation of the basic reproduction number of novel coronavirus (2019-nCoV) in China, from 2019 to 2020: a data-driven analysis in the early phase of the outbreak. Int J Infect Dis. 2020;92:214–217.

[31] Holshue ML, DeBolt C, Lindquist S, et al. First case of 2019 novel coronavirus in the United States. N Engl J Med. 2020;382(10):929–936.

[32] Núñez-Delgado A. What do we know about the SARS-CoV-2 coronavirus in the environment? Sci Total Environ. 2020;727:138647.

[33] To KK, Tsang OT, Chik-Yan Yip C, et al. Consistent detection of 2019 novel coronavirus in saliva. Clin Infect Dis. 2020. DOI:10.1093/cid/ciaa149

[34] Liu L, Wei Q, Alvarez X, et al. Epithelial cells lining salivary gland ducts are early target cells of severe acute respiratory syndrome coronavirus infection in the upper respiratory tracts of rhesus macaques. J Virol. 2011;85(8):4025–4030.

[35] Chan VW, Chiu PK, Yee CH, et al. A systematic review on COVID-19: urological manifestations, viral RNA detection and special considerations in urological conditions. World J Urol. 2020. DOI:10.1007/s00345-020-03246-4

[36] Chen H, Guo J, Wang C, et al. Clinical characteristics and intrauterine vertical transmission potential of COVID-19 infection in nine pregnant women: a retrospective review of medical records. Lancet. 2020; 395(10226):809–815.

[37] Dong L, Tian J, He S, et al. Possible vertical transmission of SARS-CoV-2 from an infected mother to her newborn. JAMA. 2020;323(18):1846–1848.

[38] Colavita F, Lapa D, Carletti F, et al. SARS-CoV-2 isolation from ocular secretions of a patient with COVID-19 in Italy with prolonged viral RNA detection. Annals Intern Med. 2020. DOI:10.7326/M20-1176

[39] Lu CW, Liu XF, Jia ZF. 2019-nCoV transmission through the ocular surface must not be ignored. Lancet. 2020;395(10224):e39.

[40] Backer JA, Klinkenberg D, Wallinga J. Incubation period of 2019 novel coronavirus (2019-nCoV) infections among travellers from Wuhan, China, 20–28 January 2020. Euro Surveill. 2020;25(5):2000062.

[41] Lauer SA, Grantz KH, Bi Q, et al. The incubation period of coronavirus disease 2019 (COVID-19) from publicly reported confirmed cases: estimation and application. Ann Intern Med. 2020;172(9):577–582.

[42] The White House. Press Briefing by Members of the President's Coronavirus Task Force. [cited 2020 Jan 31]. Available from: www.whitehouse.gov/briefings-statements/press-briefing-members-presidents-coronavirus-task-force

[43] European Centre for Disease Prevention and Control. Algorithm for the management of contacts of probable or confirmed COVID-19 cases. [cited 2020 Feb 25]. Available from: https://www.ecdc.europa.eu/en/publications-data/algorithm-management-contacts-probableor-confirmed-covid-19-cases.

[44] Corman VM, Landt O, Kaiser M, et al. Detection of 2019 novel coronavirus (2019-nCoV) by real-time RT-PCR. Euro Surveill. 2020;25(3):2000045.

[45] Reusken C, Broberg EK, Haagmans B, et al. Laboratory readiness and response for novel coronavirus (2019-nCoV) in expert laboratories in 30 EU/EEA countries. Euro Surveill. 2020;25(6):2000082.

[46] Chan JF, Yip CC, To KK, et al. Improved molecular diagnosis of COVID-19 by the novel, highly sensitive and specific COVID-19-RdRp/Hel real-time reverse transcription-polymerase chain reaction assay validated in vitro and with clinical specimens. J Clin Microbiol. 2020;58(5):pii: e0031020.

[47] Cheng PK, Wong DA, Tong LK, et al. Viral shedding patterns of coronavirus in patients with probable severe acute respiratory syndrome. Lancet. 2004; 363(9422):1699–1700.

[48] Al-Abdely HM, Midgley CM, Alkhamis AM, et al. Middle East respiratory syndrome coronavirus infection dynamics and antibody responses among clinically diverse patients, Saudi Arabia. Emerging Infect Dis. 2019;25(4):753–766.

[49] Loeffelholz MJ, and, Tang Y-W. Laboratory diagnosis of emerging human coronavirus infections – the state of the art. Emerg Microb Infect. 2020;9: 747–756.

[50] Yelin I, Aharony N, Shaer Tamar E, et al. Evaluation of COVID-19 RT-qPCR test in multi-sample pools. Clin Infect Dis. 2020. DOI:10.1093/cid/ciaa531

[51] Hogan CA, Sahoo MK, Pinsky BA. Sample pooling as a strategy to detect community tTransmission of SARS-CoV-2. JAMA. 2020;323(19):1967.

[52] Wyllie AL, Fournier J, Casanovas-Massana A, et al. Saliva is more sensitive for SARS-CoV-2 detection in COVID-19 patients than nasopharyngeal swabs. medRxiv. 2020. DOI:10.1101/2020.04.16.20067835

[53] Wu Z-S, Zhang Z-Q, Wu S. Focus on the crosstalk between COVID-19 and urogenital systems. J Urol. 2020;204(1):7–8.

[54] Guan W-J, Ni Z-Y, Hu Y, et al. Clinical characteristics of 2019 novel coronavirus infection in China. New Engl J Med. 2020;382(18):1708–1720.

[55] Notomi T, Okayama H, Masubuchi H, et al. Loop-mediated isothermal amplification of DNA. Nucleic Acids Res. 2000;28(12):E63.

[56] Broughton JP, Deng X, Yu G, et al. CRISPR–Cas12-based detection of SARS-CoV-2. Nat Biotech. 2020. DOI:10.1038/s41587-020-0513-4

[57] Liu Y, Yan L-M, Wan L, et al. Viral dynamics in mild and severe cases of COVID-19. Lancet Infect Dis. 2020;20(6):656–657.

[58] Zhao J, Yuan Q, Wang H, et al. Antibody responses to SARS-CoV-2 in patients of novel coronavirus disease 2019. Lancet. 2020. DOI:10.1093/cid/ciaa344

[59] Zhang G, Nie S, Zhang Z, et al. Longitudinal change of SARS-Cov2 antibodies in patients with COVID-19. J Infect Dis. 2020. DOI:10.1093/infdis/jiaa229

[60] Long Q-X, Liu B-Z, Deng H-J, et al. Antibody responses to SARS-CoV-2 in patients with COVID-19. Nature Med. 2020;26:845–848.

[61] Haveri A, Smura T, Kuivanen S, et al. Serological and molecular findings during SARS-CoV-2 infection: the first case study in Finland, January to February. Euro Surveill. 2020;25(11):2000266.
[62] Amanat F, Stadlbauer D, Strohmeier S, et al. A serological assay to detect SARS-CoV-2 seroconversion in humans. Nat Med. 2020. DOI:10.1101/2020.03.17.20037713
[63] Elshabrawy HA, Coughlin MM, Baker SC, et al. Human monoclonal antibodies against highly conserved HR1 and HR2 domains of the SARS-CoV spike protein are more broadly neutralizing. PLoS One. 2012;7(11):e50366.
[64] Group PIW, Multi-National P, Davey RT, Jr, et al. A randomized, controlled trial of zmapp for Ebola virus infection. N Engl J Med. 2016;375:1448–14456.
[65] Wang C, Li W, Drabek D, et al. A human monoclonal antibody blocking SARS-CoV-2 infection. Nat Commun. 2020;11(1):2251.
[66] Okba NMA, Müller MA, Li W, et al. Severe acute respiratory syndrome coronavirus 2-specific antibody responses in coronavirus disease 2019 patients. Emerg Infect Dis. 2020. DOI:10.3201/eid2607.200841
[67] Tang F, Quan Y, Xin ZT, et al. Lack of peripheral memory B cell responses in recovered patients with severe acute respiratory syndrome: a six-year follow-up study. JI. 2011;186(12):7264–7268.
[68] Foundation for Innovative New Diagnostics (FIND). 2020. Available from: https://www.finddx.org/covid-19/sarscov2-eval-immuno/
[69] Chan JF, Kok KH, Zhu Z, et al. Genomic characterization of the 2019 novel human-pathogenic coronavirus isolated from a patient with atypical pneumonia after visiting Wuhan. Emerg Microbes Infect. 2020;9(1):221–236.
[70] An overview of the rapid test situation for COVID-19 diagnosis in the EU/EEA. Technical Report, 1 April 2020. European Centre for Disease Prevention and Control. Stockholm, 2020. Available from: https://www.ecdc.europa.eu/sites/default/files/documents/Overview-rapid-test-situation-for-COVID-19-diagnosis-EU-EEA.pdf
[71] Nuccetelli M, Grelli S, Ciotti M, et al. SARS-CoV-2 infection serology: a useful tool to overcome lockdown? Cell Death Discov. 2020;6:38.
[72] Carter LJ, Garner LV, Smoot JW, et al. Assay techniques and test development for COVID-19 diagnosis. ACS Cent Sci. 2020;6(5):591–605.
[73] Huang C, Wen T, Shi F-J, et al. Rapid detection of IgM antibodies against the SARS-CoV-2 virus via colloidal gold nanoparticle-based lateral-flow assay. ACS Omega. 2020;5(21):12550–12556.
[74] Shen B, Zheng Y, Zhang X, et al. Clinical evaluation of a rapid colloidal gold immunochromatography assay for SARS-Cov-2 IgM/IgG. Am J Transl Res. 2020;12(4):1348–1354.
[75] Wrapp D, Wang N, Corbett KS, et al. Cryo-EM structure of the 2019-nCoV spike in the prefusion conformation. Science. 2020;367(6483):1260–1263.
[76] Letko M, Marzi A, Munster V. Functional assessment of cell entry and receptor usage for SARS-CoV-2 and other lineage B betacoronaviruses. Nat Microbiol. 2020;5(4):562–569.
[77] Walls AC, Park YJ, Tortorici MA, et al. Structure, function, and antigenicity of the SARS-CoV-2 spike glycoprotein. Cell. 2020;181(2):281–292.e6. pii: S0092-8674(20)30262-2.
[78] Shang J, Ye G, Shi K, et al. Structural basis of receptor recognition by SARS-CoV-2. Nature. 2020;581(7807):221–224.
[79] Hoffmann M, Kleine-Weber H, Schroeder S, et al. SARS-CoV-2 cell entry depends on ACE2 and TMPRSS2 and is blocked by a clinically proven protease inhibitor. Cell. 2020;181(2):271–280.
[80] Wang W, Su B, Pang L, et al. High-dimensional immune profiling by mass cytometry revealed immunosuppression and dysfunction of immunity in COVID-19 patients. Cell Mol Immunol. 2020;17(6):650–652.
[81] Chu H, Zhou J, Wong BH, et al. Middle East respiratory syndrome coronavirus efficiently infects human primary T lymphocytes and activates the extrinsic and intrinsic apoptosis pathways. J Infect Dis. 2016;213(6):904–914.
[82] Taylor RC, Cullen SP, Martin SJ. Apoptosis: controlled demolition at the cellular level. Nat Rev Mol Cell Biol. 2008;9(3):231–241.
[83] Youle RJ, Strasser A. The BCL-2 protein family: opposing activities that mediate cell death. Nat Rev Mol Cell Biol. 2008;9(1):47–59.
[84] Hardwick JM, Youle RJ. SnapShot: BCL-2 proteins. Cell. 2009;138(2):404, 404.e1.
[85] Wei L, Sun S, Xu CH, et al. Pathology of the thyroid in severe acute respiratory syndrome. Hum Pathol. 2007;38(1):95–102.
[86] Yeung M-L, Yao Y, Jia L, et al. MERS coronavirus induces apoptosis in kidney and lung by upregulating Smad7 and FGF2. Nat Microbiol. 2016;1(3):16004.
[87] Tan Y-X, Tan THP, Lee M-R, et al. Induction of apoptosis by the severe acute respiratory syndrome coronavirus 7a protein is dependent on its interaction with the Bcl-XL protein. J Virol. 2007;81(12):6346–6355.
[88] Tao X, Hill TE, Morimoto C, et al. Bilateral entry and release of Middle East respiratory syndrome coronavirus induces profound apoptosis of human bronchial epithelial cells. J Virol. 2013;87(17):9953–9958.
[89] Chen Y, Feng Z, Diao B, et al. The novel severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) directly decimates human spleens and lymph nodes. medRxiv. DOI:10.1101/2020.03.27.20045427
[90] Bordi L, Castilletti C, Falasca L, et al. Bcl-2 inhibits the caspase-dependent apoptosis induced by SARS-CoV without affecting virus replication kinetics. Arch Virol. 2006;151(2):369–377.
[91] Chan CM, Ma CW, Chan WY, et al. The SARS-coronavirus membrane protein induces apoptosis through modulating the Akt survival pathway. Arch Biochem Biophys. 2007;459(2):197–207.
[92] Khan S, Fielding BC, Tan TH, et al. Over-expression of severe acute respiratory syndrome coronavirus 3b protein induces both apoptosis and necrosis in Vero E6 cells. Virus Res. 2006;122(1–2):20–27.

[93] Schaecher SR, Touchette E, Schriewer J, et al. Severe acute respiratory syndrome coronavirus gene 7 products contribute to virus-induced apoptosis. J Virol. 2007;81(20):11054–11068.
[94] Tan YJ, Fielding BC, Goh PY, et al. Overexpression of 7a, a protein specifically encoded by the severe acute respiratory syndrome coronavirus, induces apoptosis via a caspase-dependent pathway. J Virol. 2004;78(24):14043–14047.
[95] Yuan X, Shan Y, Zhao Z, et al. G0/G1 arrest and apoptosis induced by SARS-CoV 3b protein in transfected cells. Virol J. 2005;2:66.
[96] Chan C-M, Tsoi H, Chan W-M, et al. The ion channel activity of the SARS-coronavirus 3a protein is linked to its pro-apoptotic function. Int J Biochem Cell Biol. 2009;41(11):2232–2239.
[97] Versteeg GA, Van De Nes PS, Bredenbeek PJ, et al. The coronavirus spike protein induces endoplasmic reticulum stress and upregulation of intracellular chemokine mRNA concentrations. J Virol. 2007;81(20):10981–10990.
[98] Feng S, Shen C, Xia N, et al. Rational use of face masks in the COVID-19 pandemic. Lancet Respir Med. 2020;8(5):434–436.
[99] Chan JF, Yuan S, Kok KH, et al. A familial cluster of pneumonia associated with the 2019 novel coronavirus indicating person-to-person transmission: a study of a family cluster. Lancet. 2020;395(10223):514–523.
[100] Zhang W, Du RH, Li B, et al. Molecular and serological investigation of 2019-nCoV infected patients: implication of multiple shedding routes. Emerg Microbes Infect. 2020;9(1):386–389.
[101] Chen W, Lan Y, Yuan X, et al. Detectable 2019-nCoV viral RNA in blood is a strong indicator for the further clinical severity. Emerg Microbes Infect. 2020;9(1):469–473.
[102] Young BE, Ong SWX, Kalimuddin S, et al. Epidemiologic features and clinical course of patients infected with SARS-CoV-2 in Singapore. JAMA. 2020;323(15):1488.
[103] van Doremalen N, Bushmaker T, Morris DH, et al. Aerosol and surface stability of SARS-CoV-2 as compared with SARS-CoV-1. New Engl J Med. 2020;382:1564–1567.
[104] Suman R, Javaid M, Haleem A, et al. Sustainability of coronavirus on different surfaces. J Clin Exp Hepatol. 2020. DOI:10.1016/j.jceh.2020.04.020
[105] Lippi G, Adeli K, Ferrari M, et al. Biosafety measures for preventing infection from COVID-19 in clinical laboratories: IFCC Taskforce Recommendations.
[106] World Health Organization. Home care for patients with COVID-19 presenting with mild symptoms and management of their contacts. Available from: https://www.who.int/publications-detail/home-care-for-patients-with-suspected-novel-coronavirus-(ncov)-infection-presenting-with-mild-symptoms-and-management-of-contacts
[107] Ahn D-G, Shin H-J, Kim M-H, et al. Current status of epidemiology, diagnosis, therapeutics, and vaccines for novel coronavirus disease 2019 (COVID-19). J Microbiol Biotechnol. 2020;30(3):313–324.
[108] Handbook of COVID-19 Prevention and Treatment. Available from: https://www.alnap.org/help-library/handbook-of-covid-19-prevention-and-treatment
[109] World Health Organization. Global surveillance for COVID-19 caused by human infection with COVID-19 virus: interim guidance. Available from: https://www.who.int/publications-detail/global-surveillance-for-human-infection-with-novel-coronavirus-(2019-nCov)
[110] Knowles SR, Phillips EJ, Dresser L, et al. Common adverse events associated with the use of ribavirin for severe acute respiratory syndrome in Canada. Clin Infect Dis. 2003;37(8):1139–1142.
[111] Wang Y, Zhang D, Du G, et al. Remdesivir in adults with severe COVID-19: a randomised, double-blind, placebo-controlled, multicentre trial. Lancet. 2020;395(10236):1569–1578. [Published correction appears in Lancet. 2020;395(10238):1694.
[112] Li HS, Kuok DIT, Cheung MC, et al. Effect of interferon alpha and cyclosporine treatment separately and in combination on Middle East respiratory syndrome coronavirus (MERS-CoV) replication in a human in-vitro and ex-vivo culture model. Antiviral Res. 2018;155:89–96.
[113] Chinese clinical guidance for COVID-19 pneumonia diagnosis and treatment (7th ed). [cited 2020 May 11]. Available from: http://kjfy.meetingchina.org/msite/news/show/cn/3337.html
[114] Xu CY, Lu SD, Ye X, et al. Combined treatment of tocilizumab and chloroquine on severe COVID-19: a case report. QJM. 2020. DOI:10.1093/qjmed/hcaa153
[115] Shen C, Wang Z, Zhao F, et al. Treatment of 5 critically ill patients with COVID-19 with convalescent plasma. JAMA. 2020;323(16):1582–1589.
[116] Prattes J, Valentin T, Hoenigl M, et al. Invasive pulmonary aspergillosis complicating COVID-19 in the ICU - a case report. Med Mycol Case Rep. 2020. DOI:10.1016/j.mmcr.2020.05.001
[117] Pham DT, Toeg H, De Paulis R, et al. Establishment and management of mechanical circulatory support during the COVID-19 pandemic. Circulation. 2020. DOI:10.1161/CIRCULATIONAHA.120.047415
[118] Gopalakrishnan A, Mossaid A, Lo KB, et al. Fulminant acute kidney injury in a young patient with novel coronavirus 2019. Cardiorenal Med. 2020. DOI:10.1159/000508179
[119] Dastan F, Saffaei A, Mortazavi SM, et al. Continues renal replacement therapy (CRRT) with disposable hemoperfusion cartridge: a promising option for severe COVID-19. J Glob Antimicrob Resist. 2020;21:340–341.
[120] Yang XH, Sun RH, Zhao MY, et al. Expert recommendations on blood purification treatment protocol for patients with severe COVID-19: recommendation and consensus. Chronic Dis Transl Med. 2020;6(2):106-114.
[121] Zhao JP, Hu Y, Du TH, et al. Expert consensus on the use of corticosteroid in patients with 2019-nCoV pneumonia. Chin J Tuber Respir Dis. 2020; 43:183–184.
[122] Jin YH, Cai L, Cheng ZS, et al. A rapid advice guideline for the diagnosis and treatment of 2019 novel coronavirus (2019-nCoV) infected pneumonia (standard version). Mil Med Res. 2020;7(1):4.

[123] Correa H, Malloy-Diniz LF, Da Silva AG. Why psychiatric treatment must not be neglected during the COVID-19 pandemic. Braz J Psychiatry. 2020. DOI:10.1590/1516-4446-2020-0995

[124] Huang YF, Bai C, He F, et al. Review on the potential action mechanisms of Chinese medicines in treating coronavirus disease 2019 (COVID-19). Pharmacol Res. 2020;158:104939.

[125] Chen Y, Liu Q, Guo D. Emerging coronaviruses: genome structure, replication, and pathogenesis. J Med Virol. 2020;92(4):418–423.

[126] FasterCures. COVID-19 treatment and vaccine tracker. Available from: https://milkeninstitute.org/covid-19-tracker

[127] Colson P, Rolain J-M, Raoult D. Chloroquine for the 2019 novel coronavirus SARS-CoV-2. Int J Antimicrob Agents. 2020;55(3):105923.

[128] Study to ask: Does antimalarial drug prevent COVID-19? Available from: https://newsroom.uw.edu/news/does-antimalarial-drug-pre-vent-covid-19-study-seeks-answers

[129] Liu T, Luo S, Libby P, et al. Cathepsin L-selective inhibitors: a potentially promising treatment for COVID-19 patients. Pharmacol Ther. 2020. DOI:10.1016/j.pharmthera.2020.107587

[130] Cheng Y, Wong R, Soo YOY, et al. Use of convalescent plasma therapy in SARS patients in Hong Kong. Eur J Clin Microbiol Infect Dis. 2005;24(1):44–46.

[131] Accorsi P, Berti P, de Angelis V, et al. Position paper on the preparation of immune plasma to be used in the treatment of patients with COVID-19. Blood Transfus. 2020;18(3):163–166.

[132] World Health Organization. Infection prevention and control during health care when novel coronavirus(nCoV) infection is suspected interim guidance. [cited 2020 Jan 25]. Available from: https://apps.who.int/iris/handle/10665/330674

[133] Thanh Le T, Andreadakis Z, Kumar A, et al. The COVID-19 vaccine development landscape. Nat Rev Drug Discov. 2020;19(5):305–306.

[134] China's development of vaccines against novel coronavirus. National Health Commission of the People's Republic of China. Available from: http://en.nhc.gov.cn/2020-03/26/c_78336.htm

[135] Ahmed SF, Quadeer AA, McKay MR. Preliminary identification of potential vaccine targets for the COVID-19 coronavirus (SARS-CoV-2) based on SARS-CoV immunological studies. Viruses. 2020;12(3):254.

[136] COVID-19 vaccine. Wikipedia. [2020 Apr 25; cited 2020 Apr 27]. Available from: https://en.wikipedia.org/w/index.php?title=COVID-19_vaccine&oldid=953116109

[137] Kim E, Erdos G, Huang S, et al. Microneedle array delivered recombinant coronavirus vaccines: immunogenicity and rapid translational development. EBioMedicine. 2020;55:102743.

[138] Nunneley DCE. COVID-19 vaccine candidates: 6 front-runners. ABC News. 2020. Available from: https://abcnews.go.com/Health/covid-19-vaccine-candidates-front-runners/story?id=69881230

[139] Symvivo. COVID-19 program vision. Available from: https://www.symvivo.com/covid-19

[140] Singh S, Kumar R, Agrawal B. Adenoviral vector-based vaccines and gene therapies: current status and future prospects. London: IntechOpen; 2018.

[141] Zhu FC, Li YH, Guan XH, et al. Safety, tolerability, and immunogenicity of a recombinant adenovirus type-5 vectored COVID-19 vaccine: a dose-escalation, open-label, non-randomised, first-in-human trial. Lancet. 2020;395(10240):1845–1854.

[142] Randolph HE, Barreiro LB. Herd immunity: understanding COVID-19. Immunity. 2020;52(5):737–741.

[143] Mo H, Zeng G, Ren X, et al. Longitudinal profile of antibodies against SARS-coronavirus in SARS patients and their clinical significance. Respirology. 2006;11(1):49–53.

[144] Callow KA, Parry HF, Sergeant M, et al. The time course of the immune response to experimental coronavirus infection of man. Epidemiol Infect. 1990;105(2):435–446.

[145] Mashamba-Thompson TP, Crayton ED. Blockchain and artificial intelligence technology for novel coronavirus disease 2019 self-testing. Diagnostics. 2020;10(4):198.

[146] McCall B. COVID-19 and artificial intelligence: protecting health-care workers and curbing the spread. Lancet Digit Health. 2020;2(4):e166–e167.

[147] Li L, Qin L, Xu Z, et al. Artificial intelligence distinguishes COVID-19 from community acquired pneumonia on chest CT. Radiology. 2020. DOI:10.1148/radiol.2020200905

[148] Rao A, Vazquez JA. Identification of COVID-19 can be quicker through artificial intelligence framework using a mobile phone-based survey in the populations when cities/towns are under quarantine. Infect Control Hosp Epidemiol. 2020;3:1–18.

[149] Beck BR, Shin B, Choi Y, et al. Predicting commercially available antiviral drugs that may act on the novel coronavirus (SARS-CoV-2) through a drug-target interaction deep learning model. Comput Struct Biotechnol J. 2020;18:784–790.

[150] Senior AW, Evans R, Jumper J, et al. Improved protein structure prediction using potentials from deep learning. Nature. 2020;577(7792):706–710.

[151] Rosenbaum L. Facing Covid-19 in Italy — ethics, logistics, and therapeutics on the epidemic's front line. N Engl J Med. 2020;382(20):1873–1875.

[152] Forsman RW. Why is the laboratory an afterthought for managed care organizations? Clin Chem. 1996;42(5):813–816.

[153] Dietzen DJ. Unleashing the power of laboratory developed tests: CLOSING gaps in COVID diagnosis and beyond. J Applied Lab Med. 2020. DOI:10.1093/jalm/jfaa077